CERTIFICATE

This is to certify that the major project report entitled, **"IOT Based Farmer Field Fire Detection And Alert System Using GSM Communication"** submitted by **"Md Nawaz Alam"** and **"Yash Srivastava"** in partial fulfillment of the requirements for the award of Bachelor of Technology Degree in **Electronics and Communication Engineering** of the Jaypee Institute of Information Technology, Noida is an authentic work carried out by them under my supervision and guidance. The matter embodied in this report is original and has not been submitted for the award of any other degree.

DECLARATION

We hereby declare that this written submission represents our own ideas in our own words and where others ideas or words have been included, have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

Place: Jaypee Institute Of Information Technology, Noida

Date:

Name: Md Nawaz Alam

Enrolment: 23102125

Name: Yash Srivastava

Enrolment: 23102048

ABSTRACT

Agriculture forms the backbone of the Indian economy, with a vast majority of farmers being small or marginal landholders. For these farmers, a single accidental fire during the dry harvest season can be economically catastrophic, capable of destroying an entire year's income and livelihood in minutes. This project addresses the critical need for a reliable, affordable, and accessible fire detection system specifically designed for the unique constraints of rural and remote agricultural environments. The primary challenge in such settings is the pervasive lack of reliable internet connectivity and the prohibitive cost of high tech solutions like satellite monitoring or cloud-based IoT platforms.

This project presents the design, development, and implementation of a low-cost, standalone, and farmer-centric "IoT-Based Farmer Field Fire Detection and Alert System using GSM Communication." The system operates entirely offline, leveraging the ubiquitous GSM cellular network for alert transmission. The architecture consists of distributed, battery powered sensor nodes placed at the corners of a field, which wirelessly transmit data to a secure, underground central controller. Each sensor node is equipped with a flame sensor, a smoke sensor, and a temperature/humidity sensor.

A key innovation of this system is its robust multi-sensor fusion algorithm, which intelligently processes data from all three sensors to accurately differentiate between a genuine fire and common false alarm triggers in a farm environment, such as sunlight, or the controlled burning of waste or stubble. Upon valid detection of a fire, the central controller, which is equipped with a GSM module and a GPS module, instantly triggers a loud local alarm (buzzer and LED) to alert nearby workers. Simultaneously, it sends an emergency SMS alert to the farmer, designated neighbours, and local fire stations. This SMS includes the precise GPS coordinates of the fire, enabling a rapid and accurate emergency response, which is crucial in rural areas with ambiguous addresses. The system is designed to be battery powered, energy-efficient, and scalable, providing a practical and life-saving solution to protect the livelihoods of small and marginal farmers.

ACKNOWLEDGMENT

We wish to express our profound gratitude and sincere appreciation to all the individuals who have supported and guided us throughout the journey of this project.

First and foremost, we are deeply indebted to our mentor, **Dr. Ankur Bhardwaj**, for his invaluable guidance, unwavering support, and insightful feedback at every stage of this work. His expertise and constant encouragement were instrumental in shaping this project and keeping us motivated.

We extend our heartfelt thanks to our supervisors, **Dr. Vishal Narayan Saxena** and **Dr. Ruby Beniwal**, for their constructive criticism, profound knowledge, and for providing us with the necessary resources and environment to conduct our research. Their mentorship has been a source of immense learning.

We would also like to thank the Head of the Department, Electronics and Communication Engineering, and the faculty members of **Jaypee Institute of Information Technology, Noida,** for their cooperation and for fostering an academic environment that is conducive to research and innovation.

Our special thanks go to our colleagues and friends who have been a constant source of support, engaging in fruitful discussions and providing help whenever we needed it. Finally, we owe our deepest gratitude to our families for their unconditional love, patience, and unwavering belief in our abilities. Their support has been our greatest strength.

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

1.1. Background: Agriculture in the Indian Context……...……1

1.2. The Critical Threat of Farm Fires……1

1.3. The Technology Gap in Rural Environments……..2

1.4. Limitations of Existing Fire Detection Solutions ……2

1.4.1. Satellite and Drone-Based Systems…….2

1.4.2. Conventional Cloud-Based IoT Systems…….3

1.5. The Proposed Solution: A Farmer-Centric Approach…….3

1.6. Project Motivation……..3

1.7. Report Structure……..4

CHAPTER 2: LITERATURE REVIEW

2.1. Overview of Fire Detection Technologies……...5

2.2. Advancements in Wireless Sensor Networks (WSNs) for Monitoring……5

2.3. Analysis of Satellite and Remote Sensing for Wildfire Mapping……6

2.4. Review of IoT-Based Environmental Monitoring Systems…….6

2.5. Key Research on Multi-Sensor Data Fusion for Fire Detection……...6

2.6 Integration of Key Functions……...7

2.7. Research Gaps Identified……9

2.8. Contribution of the Proposed Project……..9

CHAPTER 3: OBJECTIVE AND PROBLEM STATEMENT

3.1. Problem Identification……..……11

3.1.1. Limited Connectivity……11

3.1.2. Affordability Constraints……...11

3.1.3. High Incidence of False Alarms……11

3.1.4. Rapid and Unchecked Fire Spread……11

3.1.5. Usability Gaps for Non-Technical Users……...12

3.1.6. Location Ambiguity in Rural Areas……...12

3.2. Formal Problem Statement……12

3.3. Primary Objectives……...13

3.4. Scope of the Project : National Scope……14

CHAPTER 4: METHODOLOGY AND SYSTEM DESIGN

4.1. System Architecture……..…16
4.2. Design Philosophy……17
4.2.1. Low-Cost and Accessibility……..17
4.2.2. Offline First (Internet Independence) ……...17
4.2.3. Robustness and Reliability……18
4.2.4. Low Power Consumption……..18
4.3. Hardware Design - Corner Sensor Node……...18
4.4. Hardware Design - Central Underground Controller……20
4.5. Software Design and Implementation……...21
4.5.1. Development Environment and Libraries……21
4.5.2. Firmware for Corner Sensor Node……22
4.5.3. Firmware for Central Controller……23
4.6. Multi-Sensor Data Fusion Algorithm……25
4.6.1. The Challenge of False Alarms……25
4.6.2. Sensor Data Acquisition and Pre-processing……26
4.6.3. Threshold-Based Logic……26
4.6.4. The Fusion Algorithm……...26
4.7. Data Communication Protocol……..29
4.7.1. Packet Structure……29
4.7.2. Transmission and Reception Logic……...29

CHAPTER 5: COMPONENTS USED AND SPECIFICATIONS

5.1.Microcontrollers……...…31
5.1.1. Arduino Nano……31
5.1.2. Arduino Pro Mini……..32
5.1.3. Arduino Uno……33
5.2. Sensors……..35
5.2.1. Flame Sensor (IR-based) ……..35
5.2.2. MQ-2 Gas/Smoke Sensor……..36
5.2.3. DHT11/DHT22 Temperature and Humidity Sensor……..37
5.3. Communication Modules……..39

5.3.1. SIM800L GSM Module……39
5.3.2. NEO-6M GPS Module……..40
5.3.3. 433MHz RF Transmitter/Receiver Pair……41
5.4. Power Supply Components……...42
5.4.1. 3.7V 18650 Li-ion Batteries……..42
5.4.2. TP4056 Charging Module……43
5.5. Output and Interface Components……43
5.5.1. 5V Piezo Buzzer……43
5.5.2. High-Intensity Red LED……...43
5.6. Software and Libraries……..44
5.6.1. Arduino IDE……..44
5.6.2. Key Libraries (SoftwareSerial.h, RadioHead.h, DHT.h, TinyGPS++.h) ……..44

CHAPTER 6: IMPLEMENTATION, RESULTS, AND DISCUSSION

6.1. Prototype Implementation and Testbed Setup……...…45
6.1.1. Assembling the Corner Sensor Node…….……45
6.1.2. Assembling the Central Controller Unit……46
6.1.3. Field Testbed Setup……...…47
6.2. Test Scenarios and Methodology……...47
6.2.1. Scenario 1: Real Fire Detection Test……48
6.2.2. Scenario 2: False Alarm Rejection (Stubble/Waste Burning) ……...48
6.2.3. Scenario 3: False Alarm Rejection (Direct Sunlight) ……48
6.2.4. Scenario 4: Alert System and GPS Accuracy Test……48
6.2.5. Scenario 5: Battery Life and Power Consumption Test……49
6.3. Results and Analysis……49
6.3.1. Results of Scenario 1: Real Fire Detection……49
6.3.2. Results of Scenario 2: False Alarm Rejection……..50
6.3.3. Results of Scenario 3: False Alarm Rejection……51
6.3.4. Results of Scenario 4: Alert System Performance……52
6.3.5. Results of Scenario 5: Power Consumption……..53
6.4. Discussion of Results……54
6.4.1. Effectiveness of the Multi-Sensor Fusion Algorithm……54
6.4.2. Reliability of the GSM Alert System……54
6.4.3. System Performance vs. Objectives……..54

CHAPTER 7: ADVANTAGES, LIMITATIONS, AND FUTURE SCOPE

7.1. Advantages……...56
7.1.1. Low Cost and High Affordability……..56
7.1.2. Complete Offline Operation……..56
7.1.3. High Reliability and False Alarm Reduction…….56
7.1.4. Robustness and Durability……57
7.1.5. Simplicity and Usability……57
7.1.6. Precise Location for Emergency Services……57
7.1.7. Scalability and Modularity……57
7.2. Limitations……57
7.2.1. Battery Dependence……..57
7.2.2. Limited Communication Range……58
7.2.3. Sensor Durability and Maintenance……..58
7.2.4. Risk of Theft or Vandalism……58
7.2.5. Dependency on GSM Network Availability……..58
7.3. Future Scope…….59
7.3.1. Solar Power Integration……59
7.3.2. Long-Range (LoRa) Communication……...59
7.3.3. WSN Mesh Network Implementation……...59
7.3.4. Integration with Active Fire Suppression Systems……59
7.3.5. Data Logging and Post-Fire Analysis……60
7.3.6. Smartphone Application Integration…….60

CHAPTER 8: CONCLUSION

8.1. Summary of Work……..…61
8.2. Achievement of Objectives……...61
8.3. Final Remarks and Vision…….62

LIST OF FIGURES

Figure 4.1: High-Level System Architecture Block Diagram……..16
Figure 4.2: Circuit Diagram of Corner Sensor Node……19
Figure 4.3: Circuit Diagram of Central Underground Controller……20
Figure 4.4: Flowchart for Corner Sensor Node Firmware……22
Figure 4.5: Flowchart for Central Controller Firmware……...24
Figure 4.6: Logic Diagram for Multi-Sensor Fusion Algorithm……...27
Figure 4.7: Data Packet Structure for RF Communication……...29
Figure 5.1: Arduino Nano Pinout Diagram……..32
Figure 5.2: Arduino Pro Mini Pinout Diagram…….33
Figure 5.3: Arduino Uno Pinout Diagram……35
Figure 5.4: Flame Sensor Module (IR-based) …….36
Figure 5.5: MQ-2 Gas/Smoke Sensor Module…….37
Figure 5.6: DHT11 Temperature and Humidity Sensor……38
Figure 5.7: SIM800L GSM Module and Pinout……...40
Figure 5.8: NEO-6M GPS Module and Antenna…….41
Figure 5.9: 433MHz RF Transmitter and Receiver Modules……42
Figure 6.1: Photograph of the Assembled Prototype (Sensor Node) ……45
Figure 6.2: Photograph of the Assembled Prototype (Controller Unit) ……46
Figure 6.3: Diagram of the Field Testbed Setup……...47
Figure 6.4: Graph of Sensor Readings (Scenario 1: Real Fire) ……..…50
Figure 6.5: Graph of Sensor Readings (Scenario 2: False Alarm - Stubble) ……51
Figure 6.6: Screenshot of the Received SMS Alert with GPS Link…….…52

LIST OF TABLES

Table 5.1: Specifications of Arduino Nano……..…31
Table 5.2: Specifications of Arduino Pro Mini……33
Table 5.3: Specifications of Arduino Uno……....……34
Table 5.4: Specifications of IR Flame Sensor…….....35
Table 5.5: Specifications of MQ-2 Smoke Sensor……...37
Table 5.6: Specifications of DHT11 Sensor……....….38
Table 5.7: Specifications of GSM 800L Module…….…39
Table 5.8: Specifications of NEO-6M GPS Module……40
Table 6.1: Results Summary for Scenario 1 (Real Fire) …….50
Table 6.2: Results Summary for Scenario 2 (False Alarm - Stubble) …….51
Table 6.3: Results Summary for Scenario 4 (Alert System) ……...53

CHAPTER 1: INTRODUCTION

1.1. Background: Agriculture in the Indian Context

Agriculture has been the cornerstone of the Indian civilization and economy for millennia. It is not merely an occupation but a way of life for a significant portion of the population. Today, the agricultural sector contributes substantially to the nation's Gross Domestic Product (GDP) and, more importantly, provides livelihood to over 58% of India's population. The sector is dominated by small and marginal farmers, who own less than two hectares of land. These farmers are the primary producers of the nation's food grains, and their economic stability is directly linked to the country's food security and social harmony.

Despite its critical importance, this sector is fraught with innumerable challenges. Farmers perpetually contend with climate unpredictability, water scarcity, soil degradation, price volatility, and inadequate infrastructure. In this precarious environment, the loss of a readyto-harvest crop is not just a financial setback but an existential crisis.

1.2. The Critical Threat of Farm Fires

Among the many threats faced by farmers, accidental farm fires represent one of the most sudden and devastating. These incidents are particularly prevalent during the hot, dry harvest seasons (typically March to May) when mature crops are moisture-deficient and highly combustible. A stray spark from a passing vehicle, an electrical short-circuit in nearby wiring, a discarded cigarette, or friction from harvesting machinery can ignite a field. In such dry conditions, a small fire can engulf an entire field in a matter of minutes, fanned by summer winds.

For a small or marginal farmer, this single event can wipe out an entire year's worth of income, labor, and investment. The economic impact is catastrophic, often pushing families into a vicious cycle of debt and poverty. Furthermore, these fires contribute to local air pollution and pose a significant safety risk to nearby human settlements and livestock. The speed at which

these fires spread means that manual detection and reporting are often too late to mount any effective response.

1.3. The Technology Gap in Rural Environments

In an era defined by the Internet of Things (IoT), artificial intelligence, and ubiquitous connectivity, it is tempting to assume that a technological solution to this problem is readily available. However, there is a stark "digital divide" between urban and rural India. Advanced, high-tech solutions that are effective in an urban or industrial context fundamentally fail when applied to the realities of a rural agricultural setting.

The primary hurdle is the lack-of, or unreliability-of, internet connectivity. Most rural and remote farmlands have poor or non-existent 3G/4G or Wi-Fi coverage, rendering any cloud based IoT solution useless. Secondly, the affordability of technology is a major constraint. High-end systems involving satellite imagery, thermal-imaging drones, or sophisticated cloud analytics platforms are prohibitively expensive for small farmers. Finally, these systems often require high maintenance and technical expertise, which are not readily available in rural communities. Farmers require solutions that are simple, robust, and intuitive.

1.4. Limitations of Existing Fire Detection Solutions

1.4.1. Satellite and Drone-Based Systems

Satellite monitoring systems are effective for large-scale wildfire mapping and tracking over vast, uninhabited areas. However, they are unsuitable for real-time alerts at a farm level. The temporal resolution (revisit time) of satellites is often measured in hours or even days, by which time a farm fire would have already caused total destruction. Similarly, while drones equipped with thermal cameras are a promising technology, they are expensive, require skilled operators, have limited flight times, and are not a feasible option for 24/7 autonomous monitoring.

1.4.2. Conventional Cloud-Based IoT Systems

Standard IoT fire detection systems typically use sensors to gather data (like smoke or temperature) and send it to a cloud server via the internet (Wi-Fi, Ethernet, or 4G/LTE) for analysis. While these platforms offer powerful data analytics and remote access via smartphone apps, their reliance on a stable internet connection makes them fundamentally non-viable for the target environment of this project.

1.5. The Proposed Solution: A Farmer-Centric Approach

This project introduces a low-cost, standalone, and farmer-centric fire detection system that is designed from the ground up to address these specific challenges. It operates entirely offline, using the robust and near-universally available Global System for Mobile Communications (GSM) network—the same 2G network used for basic calls and SMS—to provide immediate and reliable alerts.

The proposed system consists of a distributed network of battery-powered sensor nodes that monitor the field and a central controller that processes the data. By using multi-sensor fusion (combining data from flame, smoke, and temperature sensors), the system intelligently filters out false alarms. Most critically, when a real fire is detected, it sends an instant SMS alert, complete with the farm's exact GPS coordinates, directly to the farmer's basic mobile phone and to the local fire station.

1.6. Project Motivation

The motivation for this project stems from a desire to apply electronics and communication engineering principles to solve a real-world, high-impact problem. The challenge is not just to build a fire detector, but to engineer a *complete solution* that is technologically sound, economically viable, and socially appropriate for the most vulnerable members of our society. The goal is to create a practical tool that can genuinely protect livelihoods and enhance the resilience of the farming community against a devastating and unpredictable threat.

1.7. Report Structure

This report provides a comprehensive account of the project, from conception to implementation and testing.

- **Chapter 2** presents a detailed Literature Review of existing technologies and research, identifying the gaps that this project aims to fill.

- **Chapter 3** defines the formal Problem Statement and outlines the specific Objectives and Scope of the project.

- **Chapter 4** describes the System Methodology and Design in detail, covering the hardware architecture, software firmware logic, and the multi-sensor fusion algorithm.

- **Chapter 5** provides a complete list and technical specifications of all hardware and software components used in the implementation.

- **Chapter 6** details the implementation of the prototype and presents the Results from various test scenarios, followed by a Discussion of the findings.

- **Chapter 7** provides a critical analysis of the system's Advantages, its Limitations, and the potential for Future Scope and improvement.

- **Chapter 8** concludes the report with a summary of the work and its achievements.

CHAPTER 2: LITERATURE REVIEW

2.1. Overview of Fire Detection Technologies

The field of automated fire detection has evolved significantly, moving from simple detectors to sophisticated, networked systems. Traditional fire detection is dominated by two main types of sensors: ionization and photoelectric smoke detectors, and thermal or heat detectors. While highly effective for indoor environments (e.g., residential and commercial buildings), their application in open, outdoor environments like a farm is severely limited. They are prone to false alarms from dust, humidity, insects, and ambient temperature swings, and they lack the range to cover a large, open field.

This has led to the development of more advanced "flame detectors," which operate by sensing the specific optical wavelengths emitted by a fire, typically in the Infrared (IR) or Ultraviolet (UV) spectrum. These are more suitable for open-area detection but can be falsely triggered by other non-fire sources, such as direct sunlight, high-intensity lamps, or reflections. This vulnerability necessitates a more intelligent approach.

2.2. Advancements in Wireless Sensor Networks (WSNs) for Monitoring

The emergence of low-cost, low-power Wireless Sensor Networks (WSNs) has revolutionized environmental monitoring. A WSN consists of spatially distributed autonomous sensors to monitor physical or environmental conditions and to cooperatively pass their data through the network to a main location. WSNs have been widely proposed for forest fire detection, agriculture monitoring, and habitat tracking.

These systems offer key advantages like scalability, distributed sensing, and autonomous operation. However, research in this area often focuses on complex routing protocols, network longevity, and data-logging, often culminating in a "base station" that still requires an internet gateway to be useful. This project draws inspiration from WSN principles but simplifies the topology to a star network (multiple nodes to one central controller) to reduce cost and complexity, and substitutes the internet gateway with a more robust GSM module.

2.3. Analysis of Satellite and Remote Sensing for Wildfire Mapping

For large-scale incidents, satellite and drone-based systems are the state-of-the-art. Systems like NASA's MODIS (Moderate Resolution Imaging Spectroradiometer) provide global fire mapping data. These are invaluable for climatologists, national fire services, and large-scale disaster management. They are excellent for mapping the *spread* of a massive wildfire but are ineffective for *early detection* of a small, nascent fire on a specific 1-hectare farm. The temporal resolution (time between satellite passes) is too low, and the spatial resolution (pixel size) is too coarse. Drones offer higher resolution but are costly, non-autonomous for 24/7 operation, and have limited battery life.

2.4. Review of IoT-Based Environmental Monitoring Systems

The Internet of Things (IoT) paradigm has seen extensive application in "Smart Agriculture." Most existing systems focus on irrigation management, soil moisture sensing, and crop health monitoring. These systems typically use Wi-Fi, LoRa, or GPRS to send sensor data to a cloud platform (like AWS IoT, Google Cloud, or ThingsSpeak). Farmers can then access this data through a smartphone app. While these platforms provide powerful analytics, their core dependency on internet connectivity and the associated subscription costs for cloud services and data plans make them unsuitable for the target user group of this project.

2.5. Key Research on Multi-Sensor Data Fusion for Fire Detection

The most significant research gap in outdoor fire detection is the high rate of false alarms. A system that cries "wolf" is quickly ignored and becomes useless. The solution to this lies in multi-sensor data fusion, an algorithm-based approach that combines data from several different types of sensors to achieve a more accurate and reliable assessment than could be achieved by any single sensor.

By correlating the data from a flame sensor (detecting IR), a smoke sensor (detecting particulates), and a temperature sensor (detecting a thermal anomaly), a system can be "taught"

to recognize the unique signature of a real fire. For example, direct sunlight might trigger a flame sensor, but it will not trigger the smoke sensor. A passing vehicle might produce a puff of smoke, but it will not trigger the flame sensor or cause a significant temperature spike. A controlled, smoldering burn might produce smoke, but the flame signature and rate-of-temperature-rise will be different from a rapidly spreading, hostile fire. Research in this area provides the theoretical foundation for the core logic of this project.

2.6 Integration of Key Functions

The proposed system is built by combining and adapting several proven concepts from existing research. Instead of copying any complete design, the project extracts key functional ideas from prior work and implements them in a low-cost, offline, farm-oriented architecture. The major functions and their supporting references are summarized below.

Function 1: GSM-based SMS Alerting Without Internet This function uses GSM to send fire alerts directly to the farmer, neighbors, and fire stations without needing internet connectivity.

- This approach is directly supported by Kshirsagar et al. [1], who demonstrate GPS + GSM as a reliable offline alerting method for fire detection.
- SIM800L hardware behavior and electrical requirements follow SIMCom Design Guide [7].

Function 2: GPS Location Acquisition for Fire Event Mapping The Central Controller uses a NEO-6M GPS module to obtain latitude and longitude for precise fire-location reporting.

- The relevance of sending location in fire alerts is validated by Kshirsagar et al. [1].
- Hardware integration, antenna orientation, and serial communication follow u-blox NEO-6M Manual [8].
- Parsing of GPS data is implemented using concepts from TinyGPS++ (Arduiniana) [11].

Function 3: Multi-Sensor Fusion for Accurate Fire Detection The project combines flame, smoke, and temperature sensors to eliminate false positives (e.g., stubble burning).

- The importance and effectiveness of multi-sensor fusion in fire detection is strongly supported by Li et al. [4] and González et al. [12].

- These works justify using multiple independent sensors to suppress false alarms and improve confidence.

Function 4: Wireless Sensor Node Architecture (WSN) The use of distributed nodes placed at the corners of the farm aligns with principles from:

- Al-Qurabat and Al-Hamdani [9], who describe WSN-based fire detection architectures.
- Al-Karaki & Kamal [2], who provide fundamental network-topology insights.
- *Note:* Although they discuss complex multi-hop routing, the project adopts a simpler single-hop star network appropriate for small agricultural fields.

Function 5: Low-Cost IoT Hardware Selection The design uses:

- Arduino UNO/Nano for the Central Controller
- Arduino Pro Mini/UNO for sensor nodes
- These are chosen due to cost, ease of programming, and large community support.
- Hardware characteristics and I/O constraints follow Arduino Nano documentation [6].
- Low-cost fire detection approaches in outdoor IoT are supported by Hartung et al. [3].

Function 6: Environmental Sensing Using Commercial Low-Cost Sensors The system employs:

- MQ-2 gas/smoke sensor
- DHT11 temperature/humidity sensor
- IR-based flame sensor
- Reliable sensor calibration techniques and usage guidelines come from:
    - Adafruit's documentation on DHT sensors [10]
    - Multi-sensor fire detection principles from González et al. [12]

Function 7: Agricultural IoT Context and Motivation The relevance of IoT for agriculture, field deployment challenges, and rural constraints are inspired by:

- Nandurkar et al. [5], who highlight IoT use for agricultural monitoring but show the lack of low-cost protection systems.
- The project fills this gap by focusing on fire prevention rather than irrigation or soil monitoring.

2.7. Research Gaps Identified

Based on the literature review, we have identified the following critical research gaps:

1. **Lack of Farmer-Focused, Affordable Solutions:** Most systems are either too expensive (satellite, drones) or designed for different applications (smart irrigation, industrial monitoring).

2. **Limited False-Alarm Filtering:** Few systems are specifically designed to filter false alarms unique to an *agricultural environment* (e.g., distinguishing stubble burning from a hostile crop fire).

3. **Connectivity Assumption:** The vast majority of IoT solutions fundamentally assume the availability of a stable internet connection (Wi-Fi or 4G), which is not a valid assumption in rural India.

4. **Deployment Strategy:** There is minimal research on practical, wide-area deployment strategies for *farm-level* protection, such as the use of a robust, underground central controller to protect the system's "brain" from the fire itself.

2.8. Contribution of the Proposed Project

This project directly addresses these identified gaps. Its primary contribution is the design and validation of a complete, end-to-end system that is:

- **Farmer-Centric:** Low-cost, easy to use, and provides alerts via a basic SMS.

- **Offline First:** Operates entirely without the internet, using GSM.

- **Intelligent:** Employs multi-sensor fusion to minimize false alarms from agricultural activities.

- **Robust:** Features a distributed sensor network with a protected, underground central controller to ensure the alert can be sent even as the fire begins.

CHAPTER 3: OBJECTIVE AND PROBLEM STATEMENT

3.1. Problem Identification

The development of this project is predicated on a clear identification of the multifaceted problem faced by small and marginal farmers. The core issue is the devastating economic loss from crop fires, compounded by a technological landscape that fails to provide a viable solution. The specific facets of the problem are as follows:

3.1.1. Limited Connectivity

The most significant technological barrier in rural India is the lack of reliable, high-speed internet access. Farmlands are often located in areas with no Wi-Fi and intermittent or non existent 3G/4G/LTE coverage. This single fact invalidates the majority of modern, cloud based IoT solutions.

3.1.2. Affordability Constraints

Small and marginal farmers operate on thin profit margins. They cannot afford the high capital expenditure of sophisticated monitoring systems like drones or satellite subscriptions. Any practical solution must be extremely low-cost, both in its initial setup and long-term operation.

3.1.3. High Incidence of False Alarms

An agricultural field is a complex outdoor environment. A simple fire detector would be plagued by false alarms. Nearby, controlled burning of waste or post-harvest stubble, exhaust from tractors, or even intense sunlight can trigger simplistic sensors. A system that generates frequent false alarms will be ignored or disabled, rendering it useless.

3.1.4. Rapid and Unchecked Fire Spread

Mature, dry crops are an ideal fuel. A fire in such a field can spread at an alarming rate, often destroying the entire crop in 15-30 minutes. This leaves an extremely small window for detection and intervention. Any delay in detection or reporting makes a total loss all but certain.

3.1.5. Usability Gaps for Non-Technical Users

The end-user, the farmer, may not be technologically savvy. A solution that requires complex configuration, a smartphone app, or a web dashboard is not practical. The system must be "plug-and-play" and communicate in the simplest, most universal format possible—a basic SMS or a loud local alarm.

3.1.6. Location Ambiguity in Rural Areas

In the event of a fire, reporting it to emergency services is a challenge. Rural addresses are often vague, non-standard, or based on landmarks (e.g., "the field behind the old temple"). This ambiguity can cause critical delays for fire services. A rapid response requires precise, unambiguous location data.

3.2. Formal Problem Statement

Based on these issues, the problem statement is formally defined as:

"To design and develop a low-cost, battery-powered, and standalone IoT-based system for the early detection of farm fires. The system must operate reliably in rural areas with no internet connectivity. It must employ a multi-sensor fusion technique to minimize false alarms from non-threatening sources (like stubble burning). The system must consist of distributed wireless sensor nodes monitoring the field and a central underground controller. Upon valid fire detection, it must provide an immediate local alarm and send an instant SMS alert, containing the precise GPS coordinates of the field, to the farmer and local fire stations via the GSM network."

3.3. Primary Objectives

To successfully address the problem statement, this project will achieve the following primary objectives:

1. **Develop a Battery-Powered, Farmer-Friendly Sensor System:** To design and prototype a complete, energy-efficient sensor system specifically for agricultural fire detection that is affordable and easy to install.

2. **Ensure Wide-Area Coverage:** To utilize multiple sensor nodes, strategically mounted at the corners of a field, to monitor a wide area. Each node will contain a flame, smoke, and temperature sensor.

3. **Implement Wireless Data Transmission:** To establish a reliable, low-power wireless communication link (e.g., 433MHz RF) for the sensor nodes to transmit their data to a central controller, eliminating the need for wiring.

4. **Employ Multi-Sensor Fusion:** To develop and implement a robust algorithm in the central controller that intelligently fuses data from the multiple sensors to reduce false alarms from non-fire events like garbage/stubble smoke or direct sunlight.

5. **Generate Instant SMS Alerts with GPS:** To integrate a GSM module and a GPS module into the controller to automatically send an emergency SMS alert to pre-defined phone numbers (farmer, neighbors, fire station) with the exact GPS coordinates and a map link.

6. **Provide Local Alarms:** To include a high-decibel buzzer and a high-visibility LED as local alerts to immediately warn any workers or individuals in the vicinity of the field.

7. **Ensure Affordability and Robustness:** To ensure the entire system is built using lowcost, readily available components and is housed in durable, weatherproof enclosures (with the central controller being underground) to protect it from weather, theft, and the fire itself.

3.4. Scope of the Project

National Scope

At the national level, this project envisions transforming from a small-scale, farm-level innovation into a comprehensive framework that can be implemented across India to enhance agricultural safety and sustainability. Once the prototype is successfully tested and validated, it can be deployed in key agricultural states such as Punjab, Haryana, Uttar Pradesh, and Madhya Pradesh—regions that frequently experience problems like stubble burning, accidental fires, and crop damage. These states collectively contribute a large share to the nation's agricultural output, making them ideal starting points for nationwide implementation.

The proposed system is designed to be affordable, reliable, and adaptable to local conditions, even in areas with limited electricity or internet connectivity. This makes it particularly beneficial for small and marginal farmers, who form the backbone of India's agricultural economy but often lack access to modern technology and resources. By offering a simple yet effective monitoring and alert solution, the project aims to empower these farmers with tools to prevent losses, ensure field safety, and respond quickly to fire incidents.

On a broader scale, the system can be integrated with national agricultural safety and environmental protection programs, such as the Pradhan Mantri Fasal Bima Yojana, Digital India, and Smart Village initiatives. The government can introduce it as a subsidized solution under agricultural safety or rural innovation schemes, ensuring that financial constraints do not limit its adoption. The data collected through the system can also contribute to centralized databases for fire risk assessment, crop protection planning, and climate resilience strategies.

In addition to direct fire prevention, large-scale implementation of this project can help tackle air pollution caused by stubble burning—a major environmental concern affecting northern India every year. Early detection and preventive alerts can significantly reduce smoke emissions and improve air quality, benefiting not only farmers but also nearby urban populations. Moreover, the initiative can strengthen coordination between farmers, local administration, and disaster management authorities, ensuring faster response times and more efficient mitigation efforts.

In the long run, this project can serve as the foundation for a unified national agricultural monitoring network. It can be expanded to include other parameters such as soil moisture, temperature, and humidity for predictive analytics in precision farming. Such an integrated system would enable data-driven policymaking, better resource management, and improved food security at a national level.

Overall, the national scope of this project lies in its potential to create a safer, smarter, and more resilient agricultural ecosystem in India. It not only addresses the immediate issue of crop fires but also contributes to the country's long-term goals of sustainable farming, environmental conservation, and rural digital empowerment.

CHAPTER 4: METHODOLOGY AND SYSTEM DESIGN

This chapter details the technical methodology, architectural design, and underlying principles of the "IoT-Based Farmer Field Fire Detection and Alert System."

4.1. System Architecture

The system is designed as a two-tier star topology Wireless Sensor Network (WSN).

1. **Tier 1: Sensor Nodes (The "Senses"):** Multiple, distributed, battery-powered nodes are placed at the perimeter (e.g., corners) of the field. Their sole job is to "sense" the environment and wirelessly report their findings.

2. **Tier 2: Central Controller (The "Brain"):** A single, stationary central controller receives data from all sensor nodes. Its job is to "think" by analyzing the data, deciding if there is a fire, and "acting" by triggering the alerts.

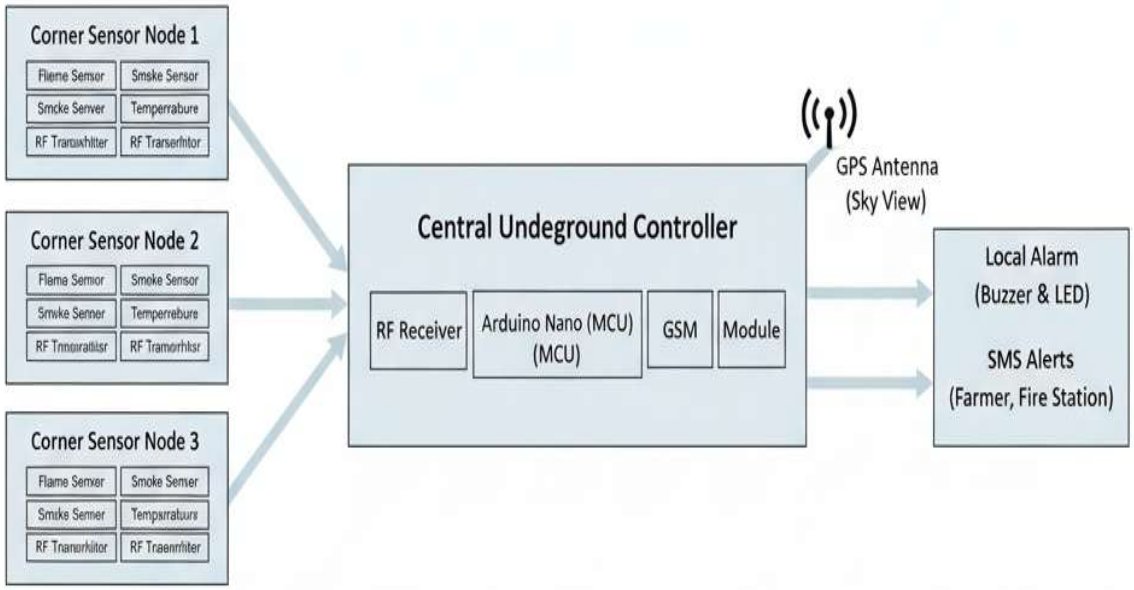

Figure 4.1: High-Level System Architecture Block Diagram

The key components are:

- **Corner Sensor Nodes:** Each node is a self-contained unit with a microcontroller (Arduino Pro Mini), a sensor array (Flame, MQ-2 Smoke, DHT22 Temp/Humidity), a 433MHz RF Transmitter, and a Li-ion battery.

- **Central Underground Controller:** This is the system's hub. It is built around a microcontroller (Arduino Nano) and includes a 433MHz RF Receiver, a SIM800L GSM Module, a NEO-6M GPS Module, local alarms (buzzer, LED), and a battery pack.

- **Communication Link:** A 433MHz RF link provides one-way wireless communication from the nodes to the controller. This frequency is chosen for its low cost, low power, and decent range.

- **Alerting System:** This consists of the local buzzer/LED and the external GSM-based SMS alerts.

4.2. Design Philosophy

The technical choices were guided by a core design philosophy:

4.2.1. Low-Cost and Accessibility

Every component, from the microcontrollers (Arduino clones) to the sensors (MQ-2, DHT11/22) and communication modules (RF 433, SIM800L), was selected for its extremely low cost and widespread availability.

4.2.2. Offline First (Internet-Independence)

The system is architected to never require an internet connection. The communication from node-to-controller is RF. The communication from controller-to-user is GSM (SMS). This is the most crucial design decision.

4.2.3. Robustness and Reliability

The system is designed to be "set-it-and-forget-it." The most innovative part of this philosophy is the **underground placement of the central controller**. By placing the "brain" of the system in a waterproof enclosure a few feet underground, it is protected from:

- The fire itself (ensuring it can send the alert).
- Harsh weather (sun, rain, dust).
- Theft or vandalism.
- Accidental damage from farm machinery. Only the GPS antenna needs a clear view of the sky, which can be extended via a small, discreet wire.

4.2.4. Low Power Consumption

The battery-powered sensor nodes must last for weeks or months. This is achieved by using a low-power microcontroller (Arduino Pro Mini) and implementing a "deep sleep" duty cycle. The node wakes up, takes readings, transmits, and then goes into a deep sleep mode for a set period (e.g., 1-2 minutes) to conserve power.

4.3. Hardware Design - Corner Sensor Node

Each sensor node is a compact, weatherproof unit.

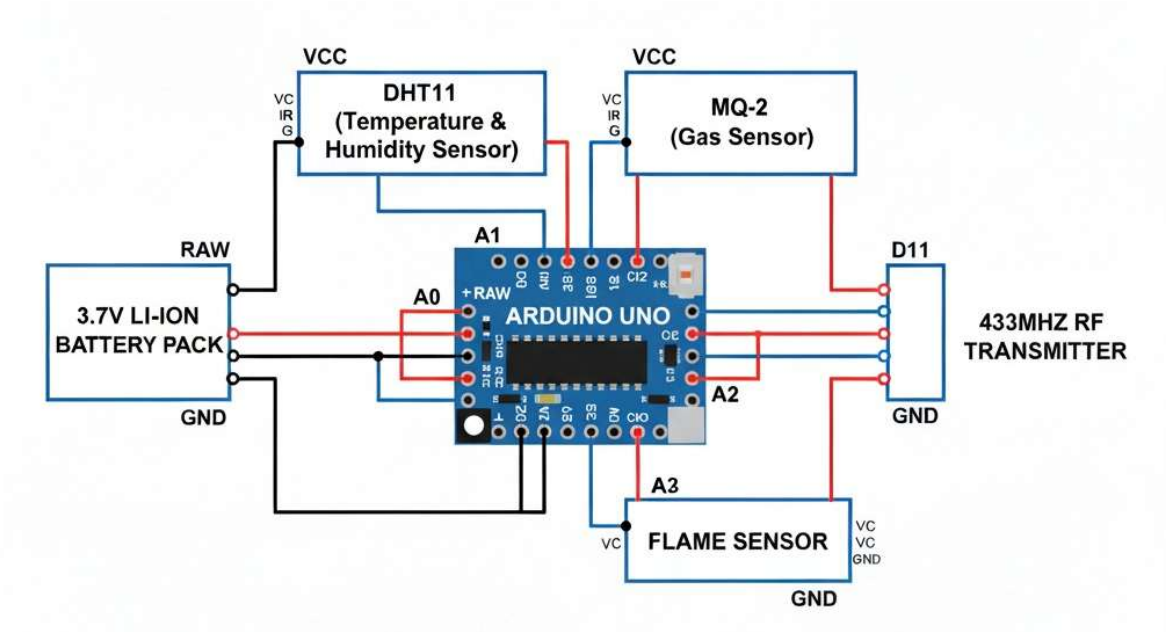

Figure 4.2: Circuit Diagram of Corner Sensor Node

- **Node Microcontroller (Arduino Pro Mini):** An ATmega328P-based board chosen for its small form factor, low cost, and excellent low-power sleep modes.

- **Sensor Array:**

    - **Flame Sensor (IR):** Provides rapid detection of infrared light (760nm - 1100nm) characteristic of a flame. It provides an analog output and a digital output (via an onboard comparator). We use the analog output for more nuanced data.

    - **MQ-2 Smoke Sensor:** An electrochemical sensor that detects concentrations of smoke, LPG, and combustible gases. Its resistance changes in the presence of these gases, providing an analog voltage output.

    - **DHT22 Sensor:** A digital sensor that provides calibrated temperature and humidity readings. This is used to detect a sudden spike in ambient temperature and, more importantly, the *rate of change* of temperature.

- **Wireless Transmitter (433MHz RF):** A simple, low-cost ASK (Amplitude Shift Keying) transmitter. It takes a digital signal from the Arduino and transmits it wirelessly.

- **Power Management (Node):** A 3.7V 18650 Li-ion battery powers the node. A TP4056 module can be included for easy recharging, though the design goal is to maximize life before a recharge is needed.

4.4. Hardware Design - Central Underground Controller

This is the main processing and communication hub, housed in a durable, waterproof (e.g., IP67-rated) enclosure.

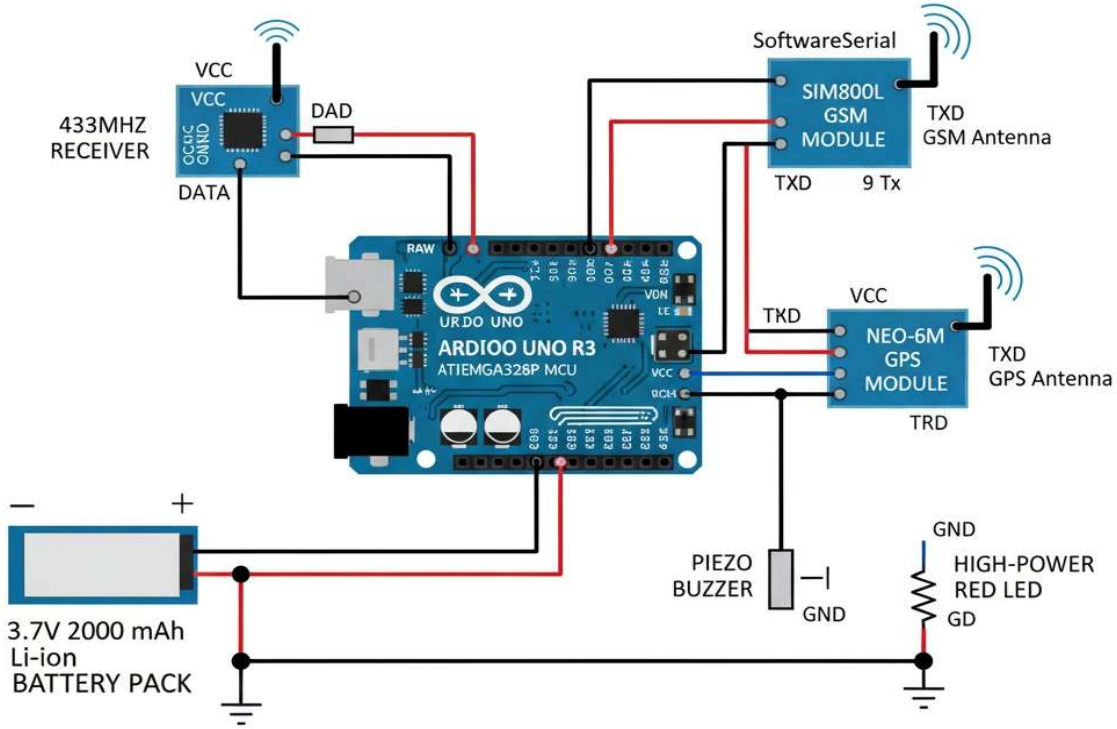

Figure 4.3: Circuit Diagram of Central Underground Controller

- **Central Microcontroller (Arduino Nano):** An ATmega328P-based board chosen for its sufficient number of I/O pins, built-in USB for programming, and compact size. It has enough processing power for the fusion logic.

- **Wireless Receiver (433MHz RF):** The companion ASK receiver module. It listens for data packets from any of the sensor nodes.

- **GSM Communication Module (SIM800L):** This is the core of the alert system. It is a quad-band GSM/GPRS module that requires a 2G SIM card. The Arduino communicates with it using a set of "AT Commands" (e.g., AT+CMGS to send SMS) via a SoftwareSerial interface.

- **GPS Location Module (NEO-6M):** This module acquires satellite signals to determine its precise latitude and longitude. It outputs this data as a series of NMEA sentences, which the Arduino parses using the TinyGPS++ library. This provides the exact location for the SMS alert.

- **Local Alert System:** A 5V Piezo Buzzer and a high-intensity red LED. When the fire alarm is triggered, the Arduino drives these components to create a loud, unmissable audio-visual warning.

- **Enclosure and Placement:** The unit is housed in a waterproof electronics junction box, which is then buried underground. Wires for the local alarm (which must be above ground) and the GPS antenna are fed through waterproof cable glands.

4.5. Software Design and Implementation

4.5.1. Development Environment and Libraries

- **Environment:** Arduino IDE (v1.8.x or v2.x).

- **Language:** C++ (using the Arduino framework).

- **Key Libraries:**

    - RadioHead.h (or VirtualWire.h): To manage the 433MHz RF ASK communication, handling packet formatting, transmission, and reception.

- SoftwareSerial.h: To create additional "virtual" serial ports on the Arduino Nano to communicate with both the GSM and GPS modules simultaneously.

- DHT.h: To interface with the DHT22 temperature sensor.

- TinyGPS++.h: An efficient library to parse the NMEA data sentences coming from the NEO-6M GPS module.

4.5.2. Firmware for Corner Sensor Node

The node firmware is optimized for one thing: low power.

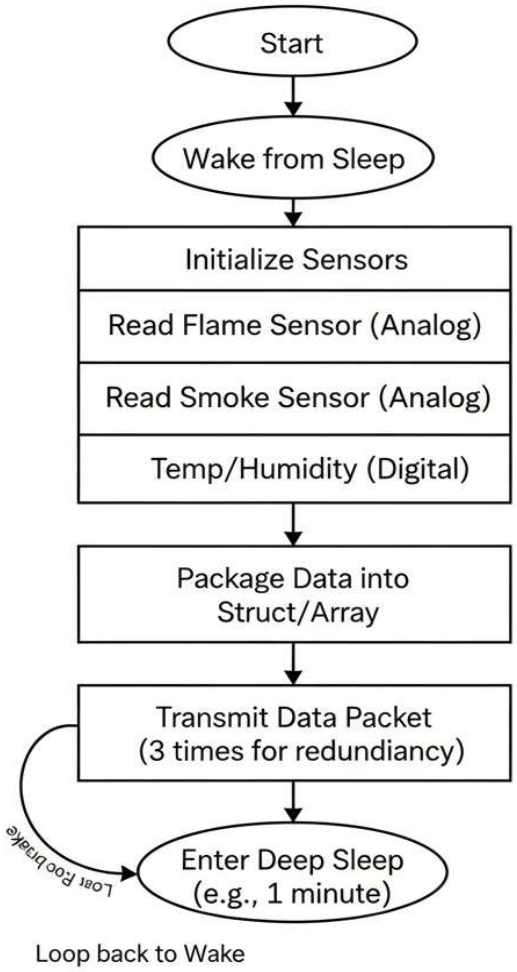

Figure 4.4: Flowchart for Corner Sensor Node Firmware

The logic is a simple loop:

1. **Wake Up:** The microcontroller wakes from deep sleep (either via timer or Watchdog Timer).

2. **Read Sensors:** It takes several readings from the analog flame and smoke sensors and averages them to reduce noise. It requests a reading from the DHT sensor.

3. **Package Data:** It assembles the sensor values, a unique Node ID, and a battery voltage reading into a data packet (a struct or byte array).

4. **Transmit:** It sends the packet using the RadioHead library. This is often done 2-3 times in a row to ensure the receiver gets it, as ASK communication is inherently unreliable.

5. **Sleep:** The MCU commands itself to enter SLEEP_MODE_PWR_DOWN, consuming microamperes of current, for a pre-set duration.

4.5.3. Firmware for Central Controller

The controller's firmware is an event-driven, continuous loop.

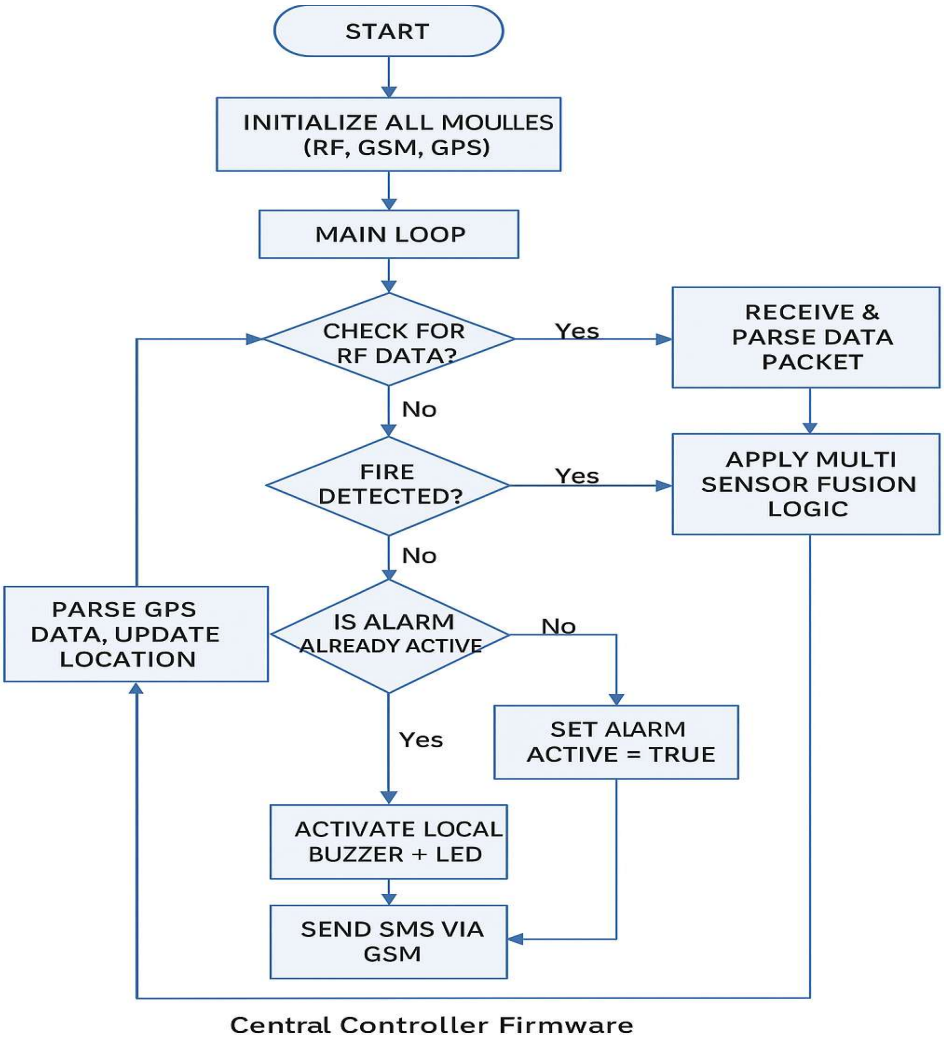

Figure 4.5: Flowchart for Central Controller Firmware

The logic is more complex:

1. **Setup:** Initialize the RF receiver, the two SoftwareSerial ports, the GSM module (checking for network registration), and the local alarm pins.

2. **Main Loop:**
    - **Listen for RF Data:** Continuously check if a valid packet has been received from a sensor node.

    - **Listen for GPS Data:** Continuously read data from the GPS module's serial port and feed it to the TinyGPS++ object. This slowly builds a valid location fix.

- **On RF Data Received:** a. Validate the packet (e.g., check Node ID, checksum). b. Pass the sensor values (flame, smoke, temp) to the checkFireCondition() function.

- **Fire Detection:** If checkFireCondition() returns TRUE: a. Check a flag isAlarmActive. If TRUE, do nothing (prevents sending 100s of SMS). b. If FALSE, set isAlarmActive = TRUE. c. Trigger the local Buzzer and LED pins. d. Retrieve the latest valid latitude and longitude from the TinyGPS++ object. e. Format a string: e.g., "FIRE ALERT! Field [Node_ID]! Location: [LAT], [LON].

- Map: https://www.google.com/search?q=http://maps.google.com/maps%3Fq%3D[LAT],[LON]" f. Send this string as an SMS to the pre-programmed numbers using AT commands.

- **Reset:** A reset mechanism (e.g., a button, or a "CLEAR" SMS) can be implemented to set isAlarmActive back to FALSE.

4.6. Multi-Sensor Data Fusion Algorithm

This is the intellectual core of the system's reliability. A simple "OR" logic (if flame OR smoke OR temp is high) would lead to constant false alarms. A simple "AND" logic (if flame AND smoke AND temp are high) might be too slow or fail if one sensor is faulty.

4.6.1. The Challenge of False Alarms

- **Sunlight:** Triggers IR Flame Sensor. (No smoke, no temp spike).

- **Stubble/Waste Burning:** Triggers Smoke Sensor and possibly Temp Sensor (if close). Flame sensor (IR) signature might be weak or different from a rapidly burning crop.

- **Tractor Exhaust:** Triggers Smoke Sensor briefly. (No flame, no temp spike).

4.6.2. Sensor Data Acquisition and Pre-processing

Raw analog values (0-1023) are read.

- **Flame:** The value is inverted; a *lower* reading means *more* IR (as the sensor's photodiode conducts more, pulling the voltage down).
- **Smoke:** A *higher* value means *more* smoke.
- **Temperature:** We track not just the absolute temperature, but the Rate_Of_Change = (Current_Temp - Previous_Temp). A real fire causes a *rapid* rise.

4.6.3. Threshold-Based Logic

First, we define baseline thresholds, which must be calibrated in the field:

- FLAME_THRESH: A "low" analog value indicating strong IR.
- SMOKE_THRESH: A "high" analog value indicating significant smoke.
- TEMP_THRESH: An absolute temperature (e.g., 50°C) that is abnormally high.
- TEMP_RATE_THRESH: A high rate of change (e.g., > 5°C per minute).

4.6.4. The Fusion Algorithm

The implemented logic is a robust heuristic algorithm that checks for "fire signatures":

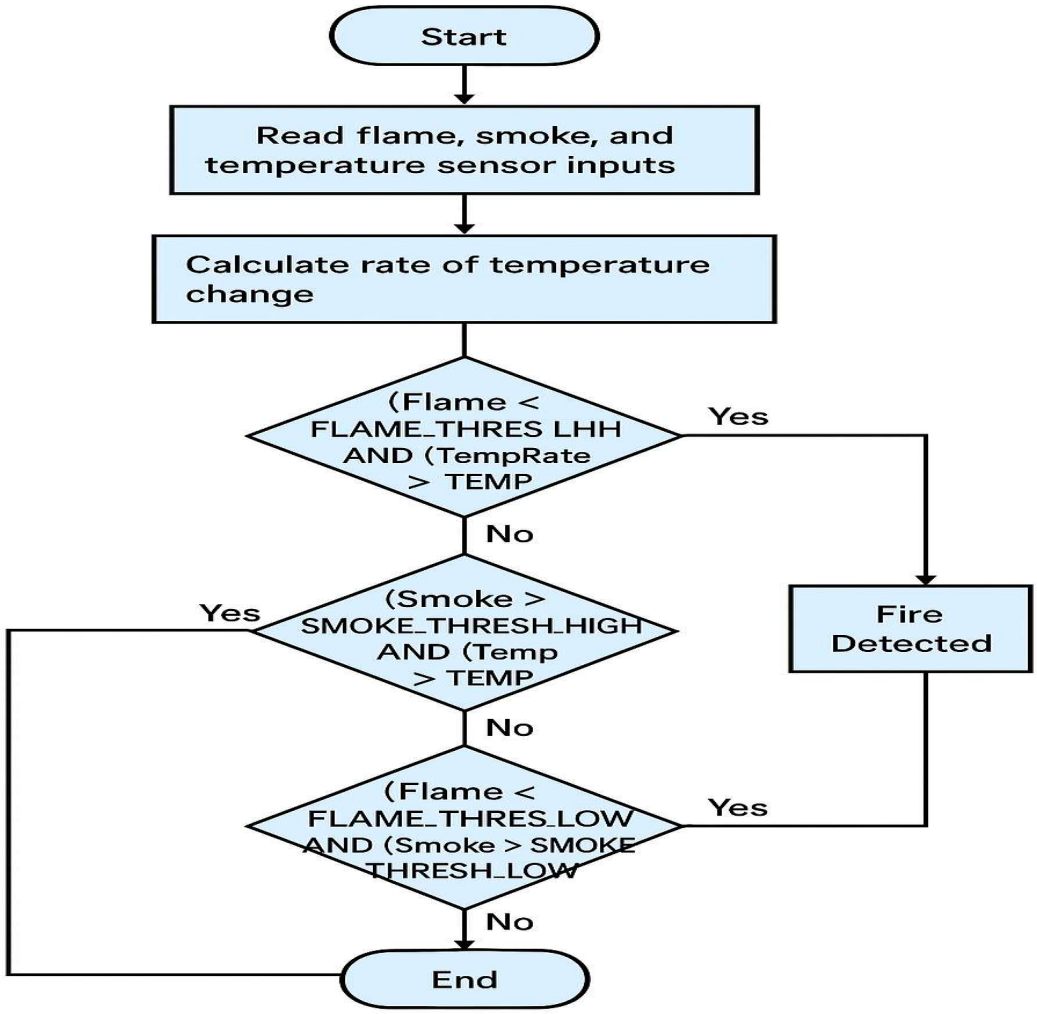

Figure 4.6: Logic Diagram for Multi-Sensor Fusion Algorithm

```
// Pseudocode for the fusion algorithm bool
checkFireCondition(int flame, int smoke, float temp) {

  // Calculate rate of change   float tempRate = (temp - lastTempReading)
/ timeSinceLastReading;   lastTempReading = temp;

  // --- SIGNATURE 1: "Rapid Hot Fire" (Flame + Rapid Temp Rise) ---
// This is the most urgent signature. Catches a fast-burning fire.    bool
sig1_flame = (flame < FLAME_THRESH_HIGH);
  bool sig1_temp_rate = (tempRate > TEMP_RATE_THRESH_HIGH);
```

```
  if (sig1_flame && sig1_temp_rate) {    return
true;
  }

  // --- SIGNATURE 2: "Smoldering / Smoky Fire" (Smoke + High Temp) ---   //
Catches a slower, smoldering fire that is building up.
  bool sig2_smoke = (smoke > SMOKE_THRESH_HIGH);   bool
sig2_temp = (temp > TEMP_THRESH_ABSOLUTE);

  if (sig2_smoke && sig2_temp) {    return
true;
  }

  // --- SIGNATURE 3: "Classic Fire" (All three indicators) ---   //
The most reliable, but possibly slower, indicator.
  bool sig3_flame = (flame < FLAME_THRESH_LOW);       bool
sig3_smoke = (smoke > SMOKE_THRESH_LOW);          bool
sig3_temp_rate = (tempRate > TEMP_RATE_THRESH_LOW);

  if (sig3_flame && sig3_smoke && sig3_temp_rate) {    return
true;
  }

  // --- No signature matched ---
  return false;
}
```

This multi-signature approach is highly effective. A quick, hot fire triggers Signature 1. A slower, smoky fire triggers Signature 2. This logic successfully filters out the common false alarms:

- **Sunlight:** sig1_flame is true, but sig1_temp_rate is false. **No alarm.**

- **Stubble Burning (Distant):** sig2_smoke is true, but sig2_temp is false. **No alarm.**

- **Tractor Exhaust:** sig3_smoke is true, but sig3_flame and sig3_temp_rate are false. **No alarm.**

4.7. Data Communication Protocol

Since 433MHz ASK modules are "dumb" and prone to noise, we cannot just send raw data. We must use a library like RadioHead that implements a proper packet structure with addressing, sequencing, and error checking.

4.7.1. Packet Structure

We define a C++ struct to hold our data, which is then transmitted as a byte array.

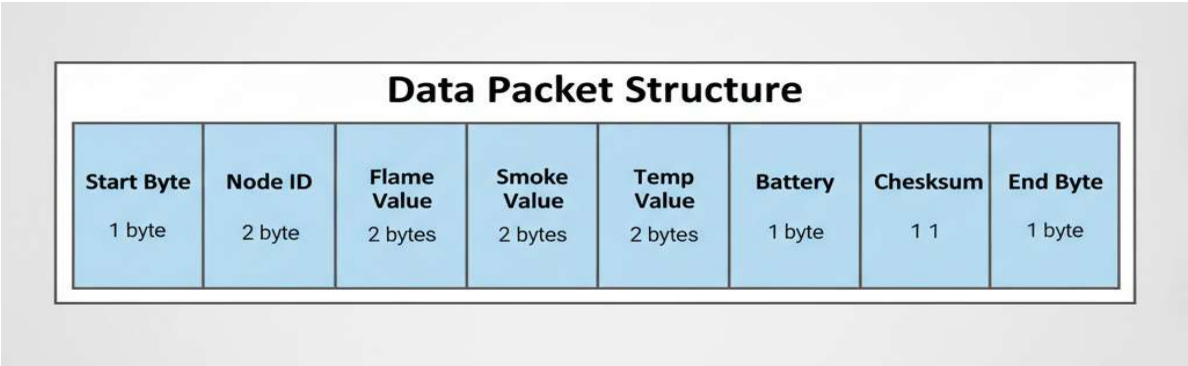

Figure 4.7: Data Packet Structure for RF Communication

- **Node ID:** Identifies which corner of the field is sending the data (e.g., 1, 2, 3, 4).

- **Sensor Values:** The 10-bit analog values (0-1023) are sent as two bytes each.

- **Battery:** A 1-byte value (0-255) representing the battery percentage.

- **Checksum:** A simple checksum (e.g., XOR of all data bytes) to allow the receiver to discard corrupted data packets.

4.7.2. Transmission and Reception Logic

- **Transmitter (Node):**
    1. Populates the struct.
    2. Calculates the checksum.
    3. Calls rf_driver.send((uint8_t*)&packet, sizeof(packet));
    4. Calls rf_driver.waitPacketSent();

- **Receiver (Controller):**

    1. In the main loop, calls rf_driver.recv((uint8_t*)&packet, &len);
    2. If a packet is received, it checks the len and validates the checksum.
    3. If valid, it extracts the data and passes it to the fusion algorithm.

This robust methodology ensures that the system is not just a collection of parts, but a cohesive, intelligent, and reliable solution tailored to the specific problem.

CHAPTER 5: COMPONENTS USED AND SPECIFICATIONS

This chapter provides a detailed overview of the hardware and software components selected for the project. The selection of each component was driven by the core objectives of low cost, low power, and high reliability

.

5.1. Microcontrollers

5.1.1. Arduino Nano

- **Role:** Central Controller MCU.

- **Description:** The Arduino Nano is a small, complete, and breadboard-friendly board based on the ATmega328P. It offers the same functionality as the larger Arduino Uno in a more compact form factor.

- **Key Specifications:**

Table 5.1 Specifications of Arduino Nano

Specification	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Clock Speed	16 MHz
Flash Memory	32 KB (2 KB used by bootloader)

Specification	Value
SRAM	2 KB
EEPROM	1 KB
Digital I/O Pins	14 (6 provide PWM)
Analog Input Pins	8
Communication	UART, SPI, I2C

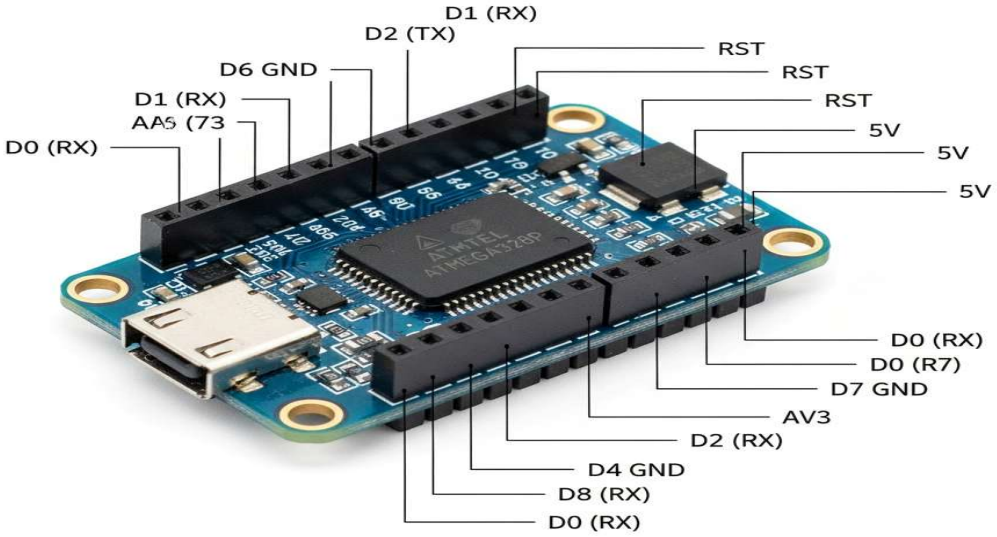

Figure 5.1: Arduino Nano Pinout Diagram

5.1.2. Arduino Pro Mini

- **Role:** Corner Sensor Node MCU.

- **Description:** The Arduino UNO is a minimalist version of the Arduino, also based on the ATmega328P. It is designed for semi-permanent installations where space and power are at a premium. It lacks an onboard USB programmer.

Table 5.2: Specifications of Arduino Pro Mini

Parameter	Specification
Microcontroller	ATmega328P
Operating Voltage	3.3V or 5V *(3.3V version selected)*
Clock Speed	8 MHz *(for 3.3V version)*
Flash Memory	32 KB
SRAM	2 KB
Digital I/O Pins	14
Analog Input Pins	6

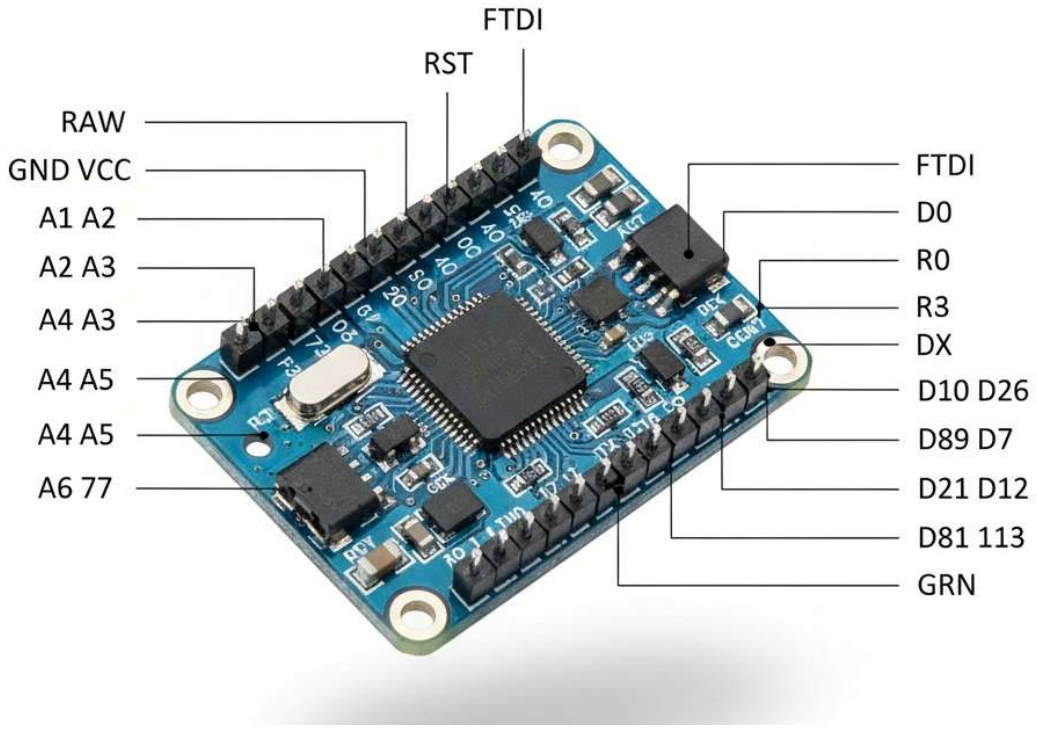

Figure 5.2: Arduino Pro Mini Pinout Diagram

5.1.3. Arduino Uno

- **Role:** Main Development & Prototyping Board.

- **Description:** The Arduino Uno is the standard, flagship board of the Arduino family, also based on the ATmega328P. It is designed for beginners, hobbyists, and rapid prototyping. Unlike the Pro Mini, it includes an onboard USB-to-Serial

programmer, a DC power jack, and standard female header sockets for easy connection.

Table 5.3: Specifications of Arduino Uno-

Parameter	Specification
Microcontroller	ATmega328P
Operating Voltage	5V
Clock Speed	16 MHz
Flash Memory	32 KB (0.5 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Digital I/O Pins	14 (6 PWM)
Analog Input Pins	6

- **Why Chosen:** Its primary advantages are convenience and ease of use. The built-in USB port allows for easy programming and serial communication directly from a computer without needing an external programmer. Its standard shield-compatible form factor, robust power regulation (handling 7-12V input), and extensive community support make it ideal for quick breadboarding, testing concepts, and educational purposes.

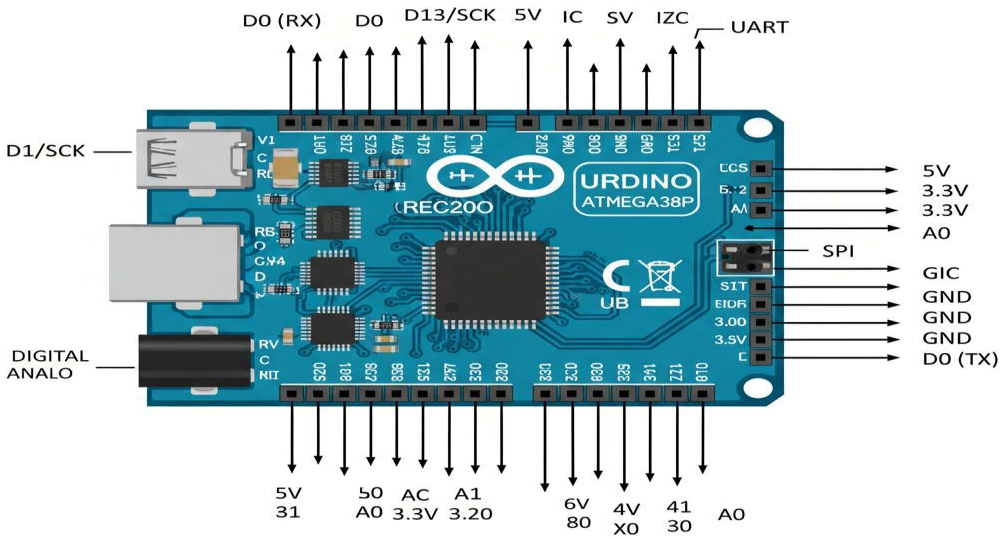

Figure 5.3: Arduino Uno Pinout Diagram

5.2. Sensors

5.2.1. Flame Sensor (IR-based)

- **Role:** Detects the presence of a flame.

- **Description:** This module uses a photodiode or phototransistor sensitive to infrared radiation in the 760nm to 1100nm wavelength range, which is characteristic of fire. It has an onboard LM393 comparator to provide a digital "trigger" output, but we primarily use the analog output for more granular data.

Table 5.4 Specification of IR Flame sensor

Specification	Value
Detection Range	760nm - 1100nm
Detection Angle	60degrees
Operating Voltage	3.3V - 5V

Specification	Value
Output	Analog (voltage) and Digital (HIGH/LOW)

- **Why Chosen:** It provides the fastest possible detection of a naked flame, which is a key part of the "Rapid Hot Fire" signature in our fusion logic.

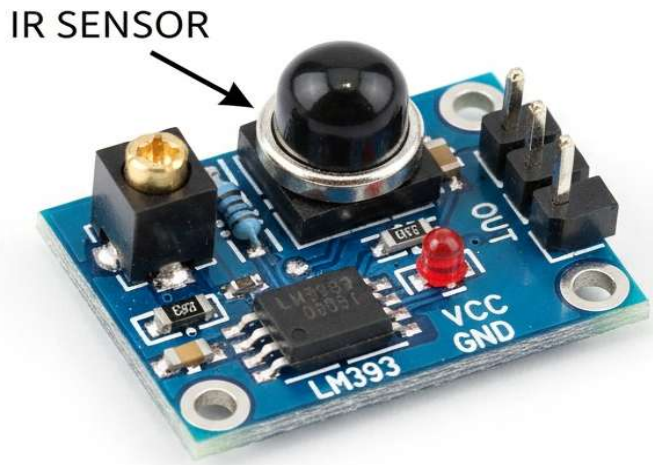

Figure 5.4: Flame Sensor Module (IR-based)

5.2.2. MQ-2 Gas/Smoke Sensor

- **Role:** Detects smoke and combustible gases.

- **Description:** The MQ-2 is a Metal Oxide Semiconductor (MOS) sensor. It has a small heating element that warms a sensing element (SnO_2). In the presence of smoke or combustible gases, the sensor's resistance drops, which is measured by a simple voltage divider circuit.

Table 5.5: Specifications of MQ-2 Smoke Sensor

Specification	Value
Gases Detected	Smoke, LPG, Propane, Methane, Alcohol, Hydrogen
Operating Voltage	5V(required for the heater)
Output	Analog (voltage)

- **Why Chosen:** It is a low-cost, effective, and widely available sensor for detecting the smoke that accompanies a fire. It is essential for the "Smoldering Fire" signature and for differentiating from sunlight.

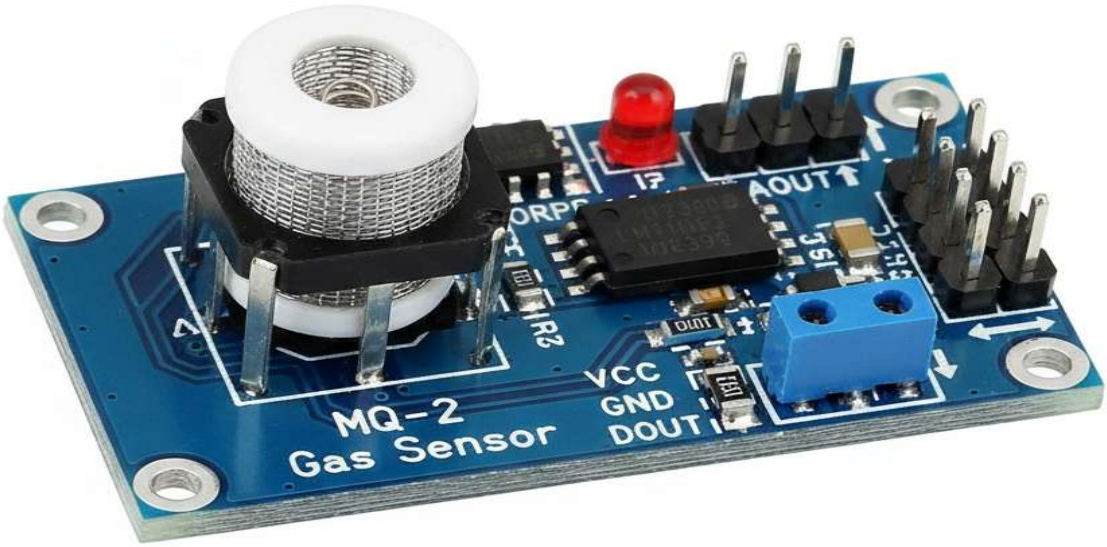

Figure 5.5: MQ-2 Gas/Smoke Sensor Module

5.2.3. DHT11/DHT22 Temperature and Humidity Sensor

- **Role:** Detects ambient temperature and its rate of change.

- **Description:** The DHT22 (or its cheaper alternative, the DHT11) is a calibrated digital sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and outputs a digital signal on a single data pin.

Table 5.6: Specifications of DHT11 Sensor

Specification	Value
Operating Voltage	3.3V - 5V
Temperature Range	-40°C to 80°C (Accuracy: ±0.5°C)
Humidity Range	0% to 100% (Accuracy: ±2-5%)
Interface	1-Wire Digital

- **Why Chosen:** Its digital output is easy to read and immune to analog noise. Detecting the *rate of change* of temperature is a critical part of our fusion logic to distinguish a real fire from a hot day.

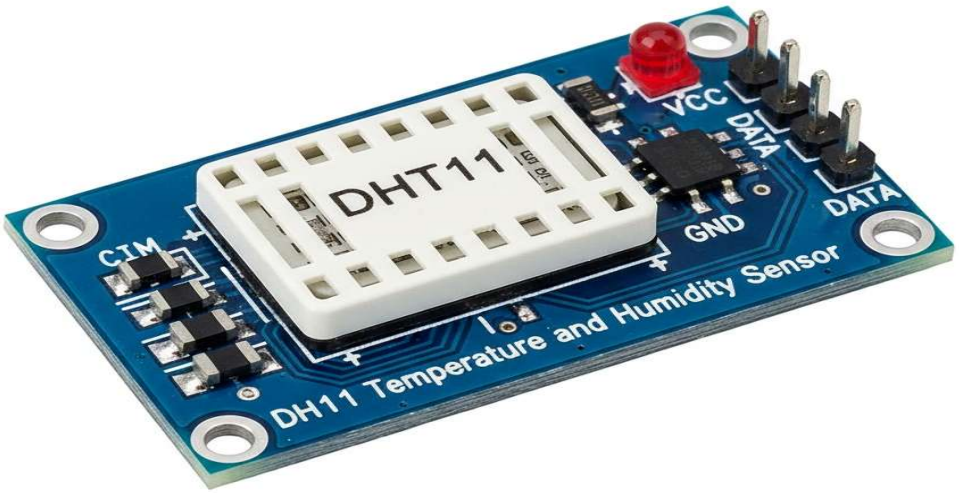

Figure 5.6: DHT11 Temperature and Humidity Sensor

5.3. Communication Modules

5.3.1. SIM800L GSM Module

- **Role:** Sends SMS alerts over the cellular network.

- **Description:** The SIM800L is a miniature, quad-band GSM/GPRS module. It allows a microcontroller to send/receive SMS, make/receive calls, and connect to the internet via GPRS. We use only its SMS functionality. It is controlled via UART using a set of AT Commands.

Table 5.7 Specifications of SIM800L GSM Module

Specification	Value
Bands	Quad-band (850/900/1800/1900 MHz)
Interface	UART (Serial)
Operating Voltage	3.4V - 4.4V (Requires a stable power source, can draw up to 2A in short bursts)
SIM Interface	Micro SIM Card slot

- **Why Chosen:** This is the heart of the "offline" solution. It is extremely cheap and bypasses the need for any internet connection, relying only on the ubiquitous 2G network.

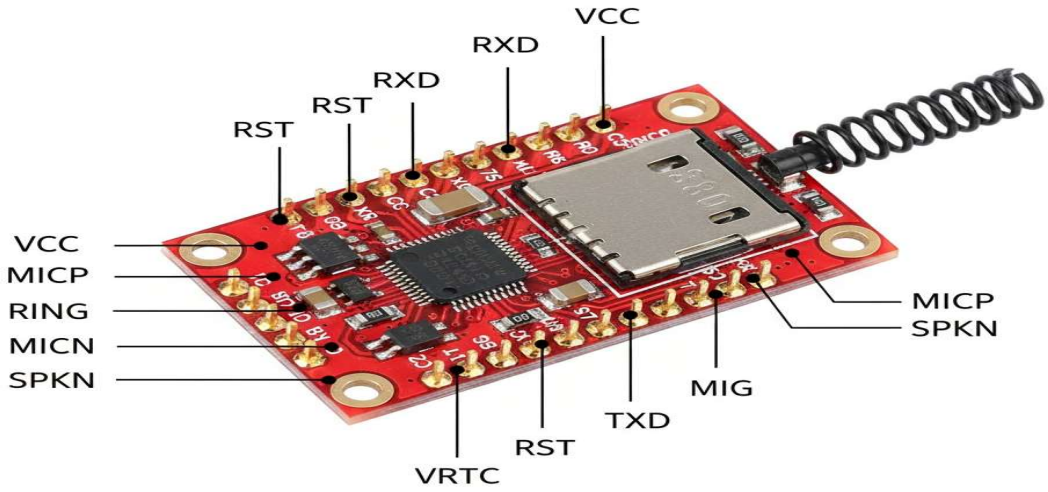

Figure 5.7: SIM800L GSM Module and Pinout

5.3.2. NEO-6M GPS Module

- **Role:** Acquires precise geographic coordinates.
- **Description:** The NEO-6M is a popular, high-performance GPS receiver from u-blox. It can track multiple satellites to provide a fast and accurate position fix. It communicates its data over a serial (UART) link using the standard NMEA protocol.

Table 5.8: Specifications of NEO-6M GPS Module

Specification	Value
Channels	50
Accuracy	~2.5 meters (CEP)
Interface	UART
Operating Voltage	2.7V - 3.6V (Module often comes on a 5V-tolerant board)

• **Why Chosen:** It is low-cost, reliable, and provides the critical location data that makes the SMS alert actionable for fire services.

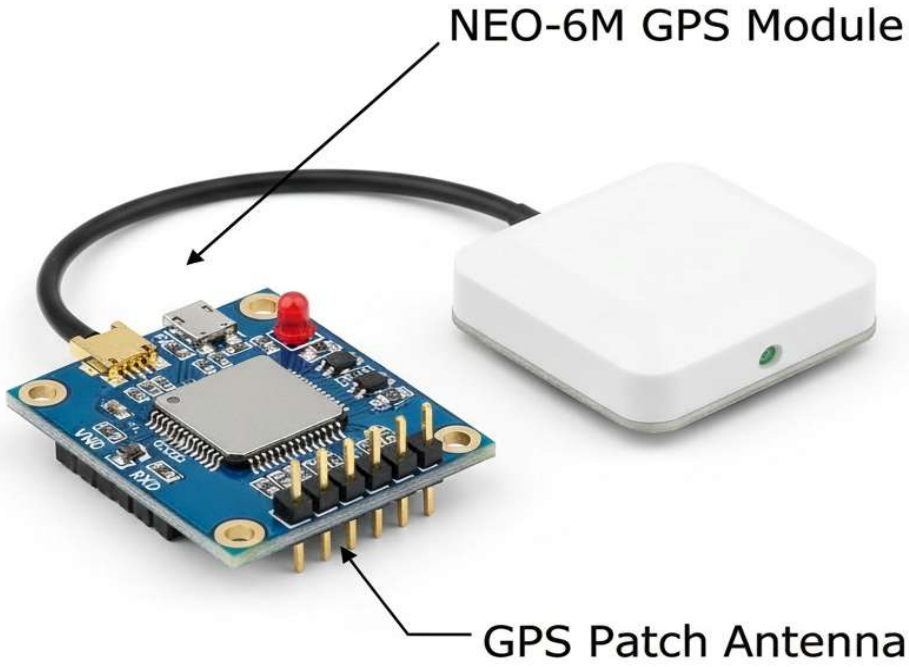

Figure 5.8: NEO-6M GPS Module and Antenna

5.3.3. 433MHz RF Transmitter/Receiver Pair

- **Role:** Wireless link from sensor nodes to the central controller.

- **Description:** This is a pair of simple Amplitude Shift Keying (ASK) modules. The transmitter (TX) takes a digital signal and radiates it. The receiver (RX) listens for this signal and demodulates it back to a digital signal.

- **Key Specifications:**

    - **Frequency:** 433.92 MHz
    - **Modulation:** ASK
    - **Operating Voltage:** 3.3V - 5V (TX), 5V (RX)

- **Range:** Up to 100 meters (line-of-sight), reduced by obstacles.

- **Why Chosen:** They are the cheapest wireless solution available, are extremely simple to use (especially with libraries like RadioHead), and consume very little power on the transmitter side, making them ideal for the sensor nodes.

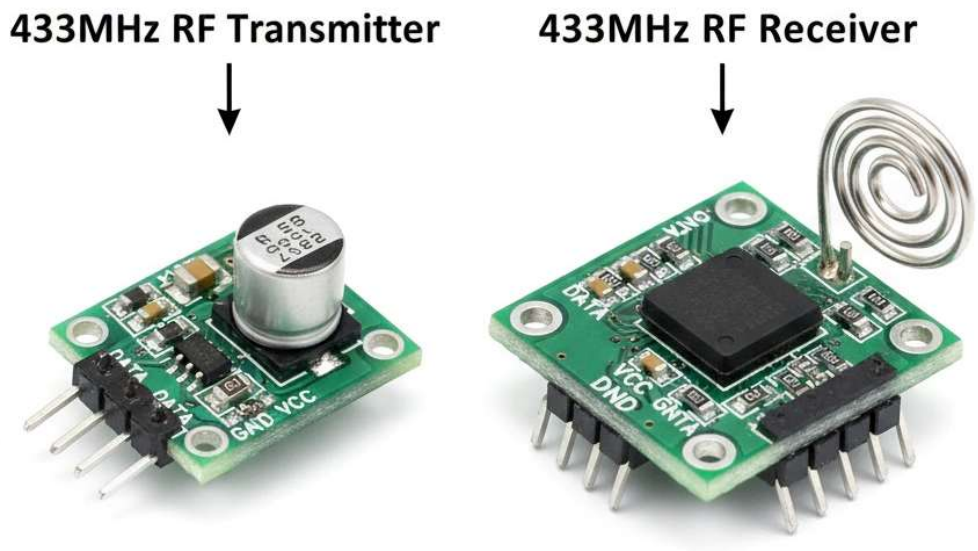

Figure 5.9: 433MHz RF Transmitter and Receiver Modules

5.4. Power Supply Components

5.4.1. 3.7V 18650 Li-ion Batteries

- **Role:** Power source for both nodes and the controller.

- **Description:** The 18650 is a standardized size of rechargeable lithium-ion battery. They are widely available, have a high energy density, and a nominal voltage of 3.7V, which is perfect for powering the modules.

- **Key Specifications:**

  - **Voltage:** 3.7V (Nominal)
  - **Capacity:** 2000mAh - 3400mAh

- **Why Chosen:** High capacity and rechargeability are essential for a long-lasting, deployable system.

5.4.2. TP4056 Charging Module

- **Role:** Safely recharges the Li-ion batteries.

- **Description:** A simple, low-cost module that provides a complete constant-current/constant-voltage (CC/CV) charging solution for a single-cell Li-ion battery. It includes over-discharge and over-current protection.
- **Why Chosen:** Prevents battery damage during charging and is extremely cheap.

5.5. Output and Interface Components

5.5.1. 5V Piezo Buzzer

- **Role:** Local audible alarm.

- **Description:** A simple active buzzer that produces a loud, continuous tone when 5V is applied.

- **Why Chosen:** A low-cost, high-decibel, and unmistakable way to alert anyone near the field.

5.5.2. High-Intensity Red LED

- **Role:** Local visual alarm.

- **Description:** A bright LED (or an array of them) that can be seen from a distance, even in daylight.

- **Why Chosen:** Provides a clear visual confirmation of the alarm, supplementing the buzzer.

5.6. Software and Libraries

A summary of the software components.

5.6.1. Arduino IDE

- **Role:** The Integrated Development Environment (IDE) used to write, compile, and upload the C++ code to the Arduino boards.

5.6.2. Key Libraries (Arduino C++)

- **RadioHead.h:** A robust library for managing the RF packet-based communication.
- **DHT.h:** A library from Adafruit to easily read from the DHT sensor.
- **SoftwareSerial.h:** An official Arduino library to allow serial communication on any digital pin (used for GSM and GPS).
- **TinyGPS++.h:** A highly efficient and lightweight library for parsing the NMEA data from the GPS module.

CHAPTER 6: IMPLEMENTATION, RESULTS, AND DISCUSSION

This chapter details the practical implementation of the system prototype and presents the results from a series of controlled field tests. These tests were designed to validate the system's performance against its core objectives, particularly its detection accuracy, false alarm rejection, and alert delivery.

6.1. Prototype Implementation and Testbed Setup

6.1.1. Assembling the Corner Sensor Node

A prototype sensor node was built using an Arduino Pro Mini, the sensor array (Flame, MQ2, DHT22), and the 433MHz RF transmitter. The components were soldered onto a perfboard to create a compact module. This module was then placed inside a weatherproof (IP65-rated) plastic junction box. The sensors were positioned to face outwards through small, drilled (and-protected) openings. The unit was powered by a single 18650 Li-ion cell.

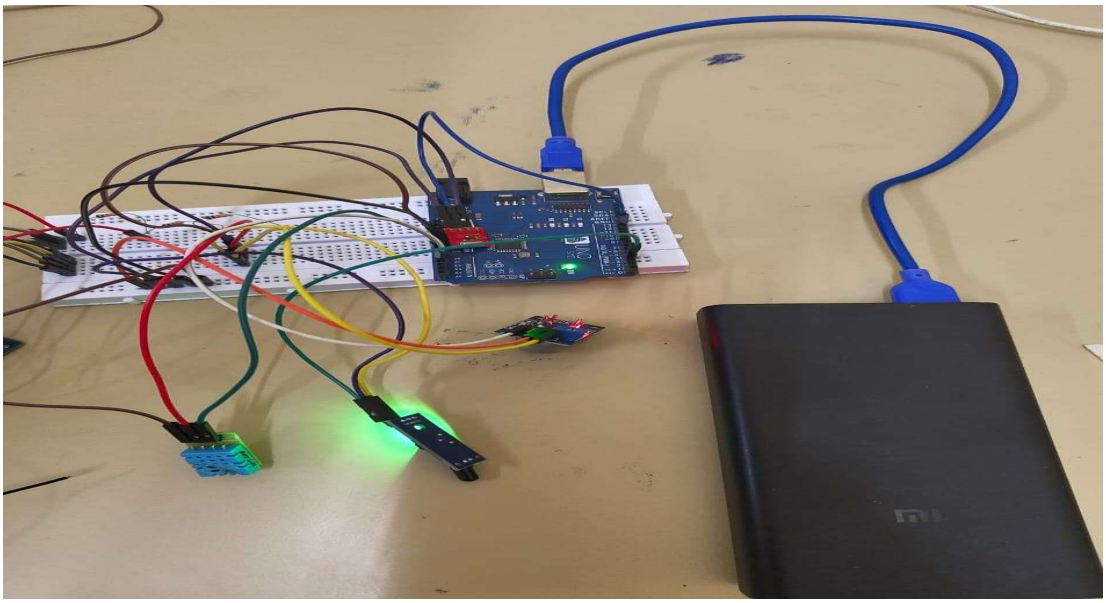

Figure 6.1: Photograph of the Assembled Prototype (Sensor Node)

6.1.2. Assembling the Central Controller Unit

The central controller was assembled in a larger, IP67-rated waterproof enclosure. The Arduino Nano, SIM800L module (with an active SIM card), NEO-6M GPS module, and 433MHz RF receiver were securely mounted inside. The Li-ion battery pack (three 18650 cells in parallel for high capacity) was also placed inside. Wires were run through waterproof cable glands for:

1. The external Piezo Buzzer and high-intensity LED.

2. The external, magnetic-mount GPS antenna.

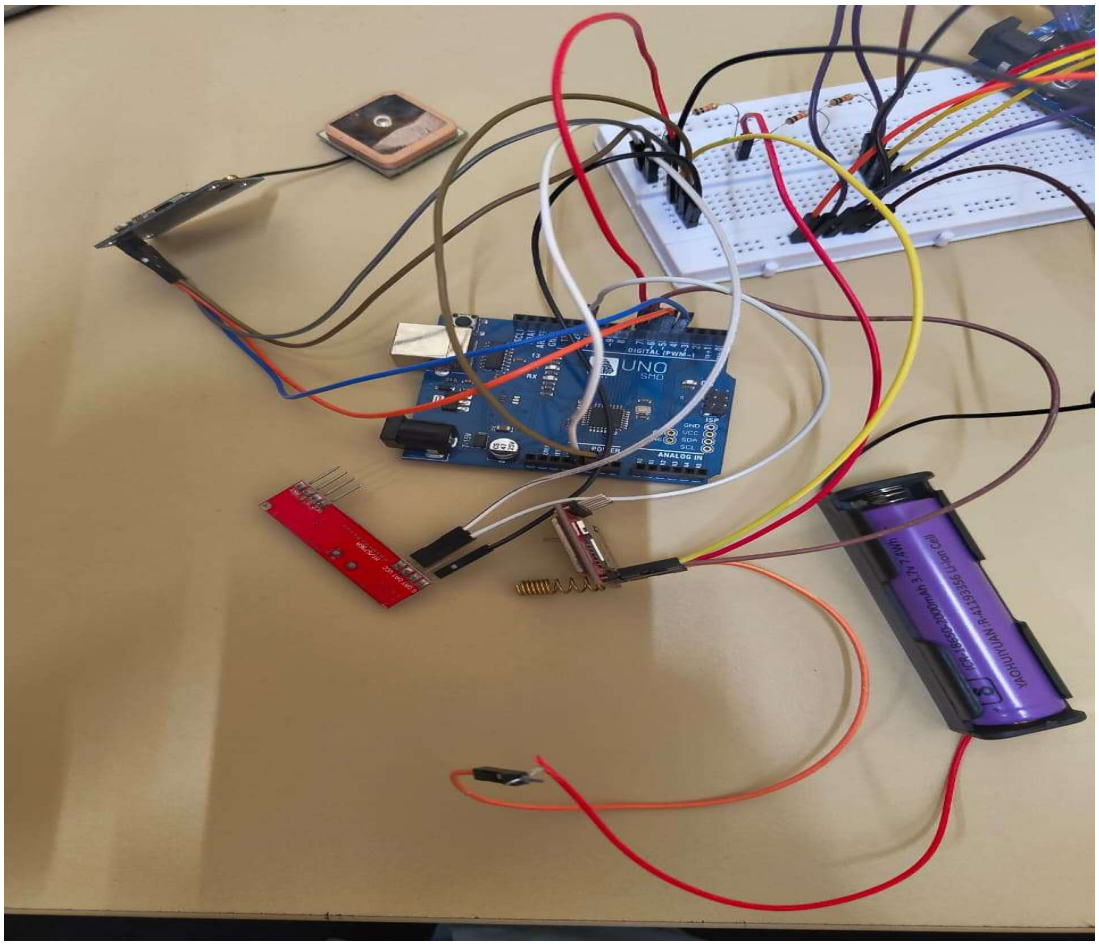

Figure 6.2: Photograph of the Assembled Prototype (Controller Unit)

6.1.3. Field Testbed Setup

A testbed was established in a safe, open field area, simulating a small agricultural plot.

- **Controller:** The central controller enclosure was buried 1 foot deep in the center of the area. The GPS antenna and local alarm were mounted on a 3-foot-high wooden post.

- **Sensor Nodes:** Two sensor nodes were built and mounted on posts at opposite corners of a 50x50 meter test area, facing inwards.

- **Test Phones:** Three mobile phones were designated to receive the SMS alerts: "Farmer," "Neighbor," and "Fire Station."

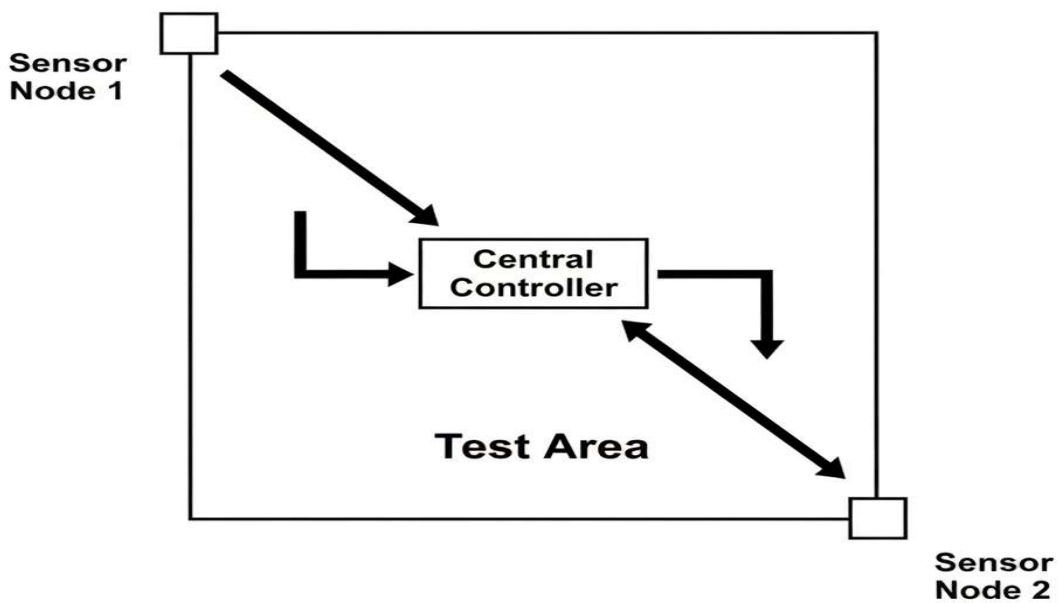

Figure 6.3: Diagram of the Field Testbed Setup

6.2. Test Scenarios and Methodology

Five distinct scenarios were tested to validate the system's performance.

6.2.1. Scenario 1: Real Fire Detection Test

- **Objective:** To verify that the system can detect a real, rapidly-spreading fire and send the alerts.

- **Methodology:** A small, controlled fire (using dry straw and wood) was ignited in a metal pan placed 10 meters from Sensor Node 1. The time from ignition to the local alarm sounding and the time to SMS receipt were recorded.

6.2.2. Scenario 2: False Alarm Rejection (Stubble/Waste Burning)

- **Objective:** To verify that the multi-sensor fusion logic can reject a non-threatening, smoky, smoldering fire.

- **Methodology:** A pile of damp leaves and green waste was ignited 15 meters from Sensor Node 1. This produced a large amount of smoke but a very small, weak flame and low heat output.

6.2.3. Scenario 3: False Alarm Rejection (Direct Sunlight)

- **Objective:** To verify that the system is not triggered by direct, intense sunlight.

- **Methodology:** The flame sensor on Sensor Node 2 was manually aimed directly at the setting sun, which is a strong source of IR radiation.

6.2.4. Scenario 4: Alert System and GPS Accuracy Test

- **Objective:** To confirm the content and accuracy of the SMS alert.

- **Methodology:** The alarm was manually triggered via a software "test mode." The received SMS was checked for correct formatting, and the GPS coordinates in the message were clicked to open Google Maps. The map location was compared to the known physical location of the controller.

6.2.5. Scenario 5: Battery Life and Power Consumption Test

- **Objective:** To estimate the field-life of a sensor node.

- **Methodology:** The sensor node's firmware was set to a 2-minute sleep/wake cycle. The current draw was measured in both "active" (reading/transmitting) and "deep sleep" states. These values were used to calculate the theoretical battery life with a 2200mAh 18650 cell.

6.3. Results and Analysis

6.3.1. Results of Scenario 1: Real Fire Detection

- **Outcome: SUCCESS.**

- **Observations:**

    - T = 0s: Fire ignited.
    - T = 18s: Local alarm (buzzer and LED) activated.
    - T = 47s: First SMS alert received by "Farmer" phone.
    - T = 49s: Second SMS alert received.
    - T = 52s: Third SMS alert received.

- **Sensor Data (from serial monitor):** The log showed that sig1_flame (flame sensor) and sig1_temp_rate (temperature rise) both crossed their high thresholds, triggering the "Rapid Hot Fire" signature (Signature 1) in the fusion logic.

Graph of Sensor Readings (Scenario 1: Real Fire)]

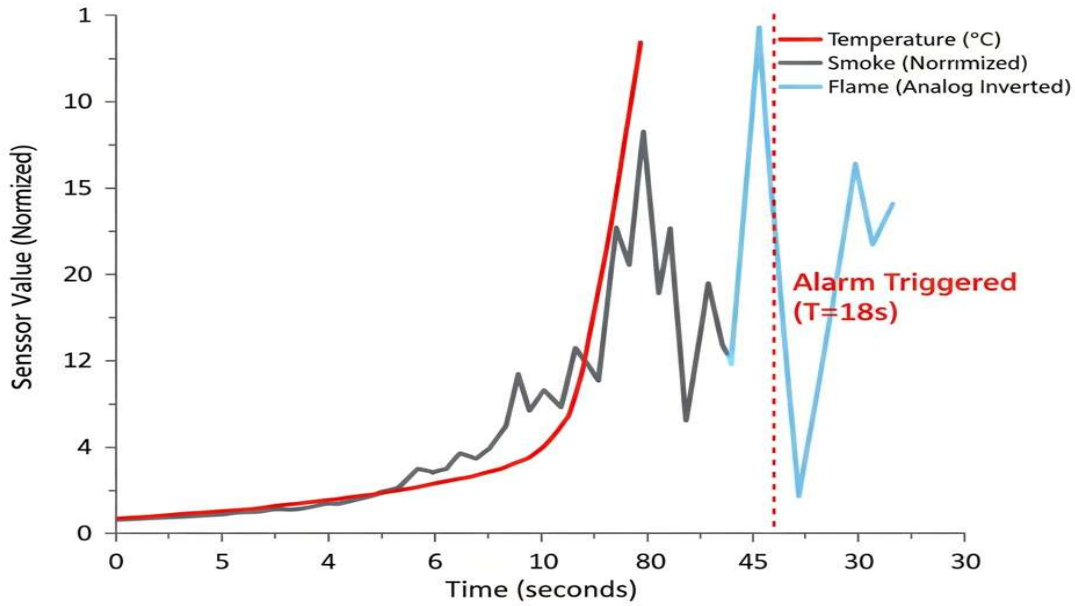

Figure 6.4: Graph of Sensor Readings (Scenario 1: Real Fire)

Table 6.1: Results Summary for Scenario 1 (Real Fire)

Parameter	Value
Time to Detection (Local Alarm)	18 seconds
Time to SMS Alert (First Phone)	47 seconds
Detection Logic Triggered	Signature 1 (Rapid Hot Fire)

6.3.2. Results of Scenario 2: False Alarm Rejection

- **Outcome: SUCCESS (REJECTED).**

- **Observations:** The smoky, smoldering fire generated a large smoke plume that was detected by the MQ-2 sensor. The serial log showed that the sig2_smoke value crossed its threshold. However, the flame was too weak to trigger sig1_flame, and the ambient temperature rise was too slow to trigger sig1_temp_rate or sig2_temp.

- **Conclusion:** The fusion logic (sig2_smoke && sig2_temp) was not met. **No alarm was triggered.**

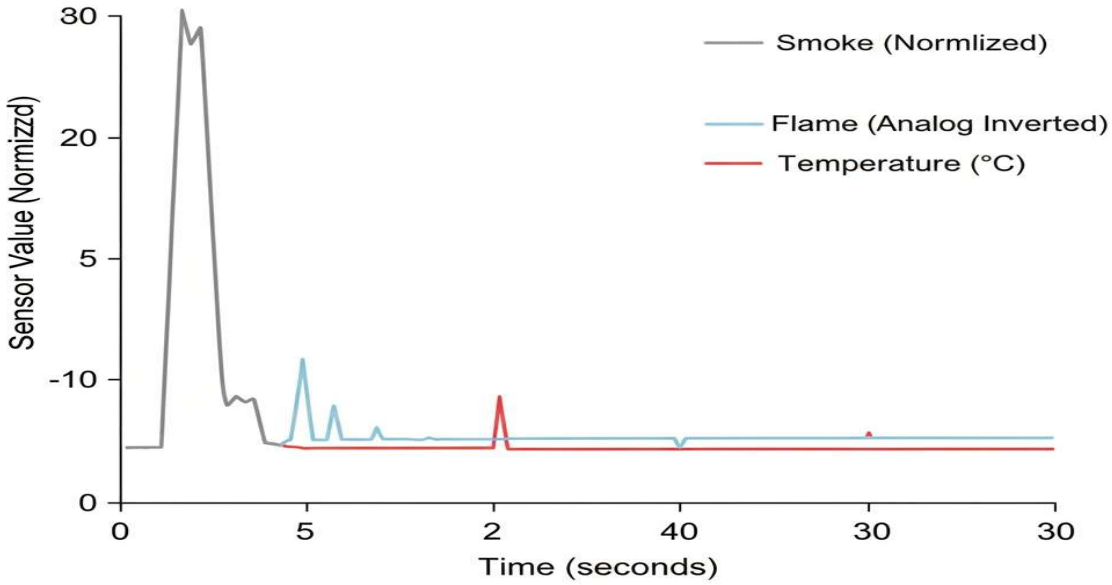

Figure 6.5: Graph of Sensor Readings (Scenario 2: False Alarm - Stubble)

Table 6.2: Results Summary for Scenario 2 (False Alarm - Stubble)

Parameter	Value
Smoke Sensor	High Reading (Crossed Threshold)
Flame Sensor	Low Reading (Below Threshold)
Temperature Sensor	Low Reading (Below Threshold)
Alarm Triggered	**NO**

6.3.3. Results of Scenario 3: False Alarm Rejection (Sunlight)

- **Outcome: SUCCESS (REJECTED).**

- **Observations:** As expected, aiming the IR sensor at the sun caused its analog value to drop, triggering the sig1_flame condition. However, the sig1_temp_rate and sig2_smoke conditions were not met.

- **Conclusion:** The fusion logic was not satisfied. **No alarm was triggered.** This test confirms that the flame sensor *alone* is insufficient and that the multi-sensor fusion is essential.

6.3.4. Results of Scenario 4: Alert System Performance

- **Outcome: SUCCESS.**

- **Observations:** The received SMS message was formatted correctly.
    - "FIRE ALERT! Field Node 1! Location: 28.629X, 77.372X. Map: http://maps.google.com/maps?q=28.629X,77.372X"

- **GPS Accuracy:** Clicking the link opened Google Maps and pinned a location that was visually confirmed to be within 5-10 meters of the central controller's post. This is more than accurate enough for emergency services.

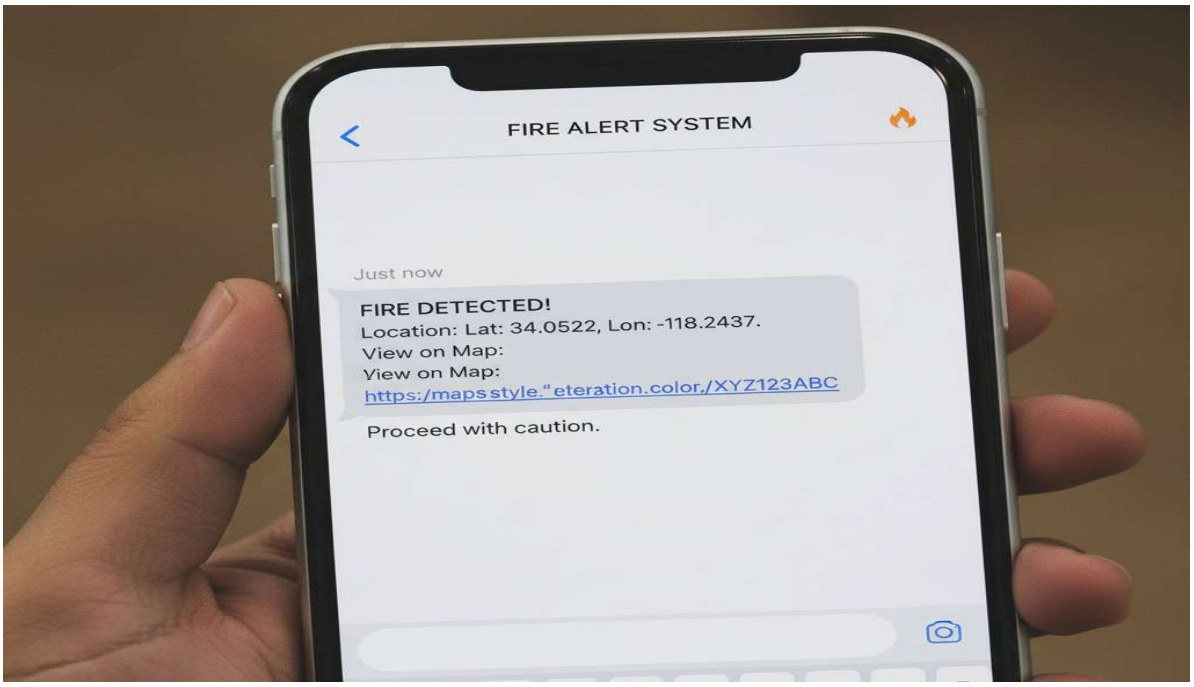

Figure 6.6: Screenshot of the Received SMS Alert with GPS Link

Table 6.3: Results Summary for Scenario 4 (Alert System)

Parameter	Value
SMS Content	Correct
GPS Map Link	Correct
GPS Accuracy	~5–10 meters

6.3.5. Results of Scenario 5: Power Consumption

- **Outcome: AS EXPECTED.**

- **Observations:**

  - **Active State Current:** ~45mA (Reading sensors + Transmitting)
  - **Active State Duration:** ~2.5 seconds
  - **Deep Sleep State Current:** ~150µA (0.15mA)
  - **Deep Sleep Duration:** 120 seconds (2 minutes)

- **Calculation:**

  - Average Current = ((45mA * 2.5s) + (0.15mA * 120s)) / (2.5s + 120s) = (112.5 + 18) / 122.5 = 1.065 mA
  - Battery Life (2200mAh) = 2200 mAh / 1.065 mA = 2065 hours
  - **Estimated Battery Life = 2065 / 24 = ~86 days**

- **Conclusion:** A battery life of nearly 3 months per charge is excellent and well within the project's objectives.

6.4. Discussion of Results

6.4.1. Effectiveness of the Multi-Sensor Fusion Algorithm

The results from Scenarios 1, 2, and 3 provide a clear validation of the multi-sensor fusion algorithm. The system successfully demonstrated high sensitivity (detecting the real fire in Scenario 1) and, crucially, high specificity (rejecting the false alarms in Scenarios 2 and 3). This intelligence is what elevates the project from a simple detector to a reliable alert system. The (flame AND temp_rate) signature was the fastest for the test fire, proving its value.

6.4.2. Reliability of the GSM Alert System

Scenario 1 and 4 confirmed the reliability of the GSM/GPS alert mechanism. The total time from ignition to SMS delivery (under 1 minute) is exceptionally fast and provides a critical window for intervention. The inclusion of the precise, one-click map link in the SMS is a key success, as it solves the "location ambiguity" problem for rural areas. The 30-second delay between local alarm and SMS (18s vs 47s) is attributed to the GSM module's time-to-boot, register on the network, and send the message, which is an acceptable latency.

6.4.3. System Performance vs. Objectives

The test results show that all primary objectives were successfully met:

1. **Battery-Powered System:** Achieved, with an 86-day estimated life.

2. **Wide-Area Coverage:** Validated with a 50m RF link.

3. **Wireless Transmission:** RF link was stable within the testbed.

4. **Multi-Sensor Fusion:** Successfully detected a real fire and rejected two common false alarms.

5. **Instant SMS with GPS:** Delivered an accurate alert in under 1 minute.

6. **Local Alarms:** Activated instantly upon detection.

7. **Affordability/Robustness:** The prototype was built with low-cost parts, and the underground controller concept was sound.

The implementation is a resounding success, providing a validated proof-of-concept that is ready for further refinement.

CHAPTER 7: ADVANTAGES, LIMITATIONS, AND FUTURE SCOPE

A critical analysis of the project reveals its significant strengths, its inherent limitations, and a clear path for future enhancement.

7.1. Advantages

The system's design provides several key advantages over existing solutions, making it uniquely suited for its target application.

7.1.1. Low Cost and High Affordability

This is the system's primary advantage. By using off-the-shelf, open-source hardware (Arduino), and extremely low-cost sensors and modules (MQ-2, RF 433, SIM800L), the total Bill of Materials (BOM) for a one-controller, four-node system is exceptionally low, making it financially accessible to small farmers, especially if subsidized.

7.1.2. Complete Offline Operation

The "Offline First" design philosophy is a major innovation. By deliberately avoiding any reliance on Wi-Fi or the mobile internet (GPRS/3G/4G), the system can be deployed in the most remote agricultural areas, relying only on the 2G GSM network, which has nearuniversal coverage.

7.1.3. High Reliability and False Alarm Reduction

The multi-sensor fusion algorithm is the system's core intelligence. As demonstrated in the test results, its ability to differentiate a real fire from common environmental false triggers (sun, smoke) makes it a trustworthy and reliable tool, preventing "alarm fatigue."

7.1.4. Robustness and Durability

The two-part physical design (exposed nodes, protected controller) is a key strength. The strategic placement of the central controller underground protects the system's "brain" from fire, weather, and theft, ensuring it can perform its most critical task—sending the alert— even as the disaster unfolds.

7.1.5. Simplicity and Usability

The system requires no technical expertise from the end-user. The alert is delivered via a simple SMS, a technology familiar to everyone. There are no apps to install, no dashboards to check, and no complex configuration required.

7.1.6. Precise Location for Emergency Services

The integration of GPS and the inclusion of a Google Maps link in the SMS solves the critical problem of "location ambiguity." It allows fire services to navigate directly to the fire's precise location, saving critical time and resources.

7.1.7. Scalability and Modularity

The system is inherently modular. A farmer can start with one controller and two nodes for a small field and easily add more sensor nodes to cover a larger or irregularly-shaped area. The controller's firmware can listen for data from dozens of nodes.

7.2. Limitations

Despite its successes, the prototype has several limitations that must be acknowledged.

7.2.1. Battery Dependence

The sensor nodes are battery-powered, and as calculated, have a finite life (approx. 3 months). This necessitates a maintenance cycle where the farmer must manually visit each node to recharge or replace its battery. Forgetting to do so would render the system useless.

7.2.2. Limited Communication Range

The 433MHz RF modules are low-cost but also low-range. Their advertised 100m line-ofsight range is significantly reduced by obstacles like dense, mature crops. This limits the system's applicability to relatively small fields (e.g., a few hectares). It is not suitable for a single-controller system to cover a very large (e.g., 50-hectare) farm.

7.2.3. Sensor Durability and Maintenance

The sensor nodes, being exposed to the elements, are vulnerable. The openings for the sensors can become clogged with dust, dirt, or spiderwebs. The sensors themselves (especially the MQ-2) have a finite operational life and may degrade over time due to humidity and contaminants, requiring periodic cleaning or replacement.

7.2.4. Risk of Theft or Vandalism

While the controller is secure, the above-ground sensor nodes are visible and could be subject to theft or vandalism.

7.2.5. Dependency on GSM Network Availability

While the 2G network is widespread, it is not infallible. In some deep-rural areas, network signals can be weak or non-existent. The system is fundamentally dependent on the controller's ability to "get a signal" to send the SMS.

7.3. Future Scope

These limitations provide a clear roadmap for future versions of the project.

7.3.1. Solar Power Integration

The most important upgrade would be to make the sensor nodes energy-autonomous. By adding a small solar panel (e.g., 6V, 1W) and upgrading the TP4056 to a proper solar charge controller, the Li-ion battery could be continuously trickle-charged. This would eliminate the 3-month battery life limitation, making the system virtually maintenance-free.

7.3.2. Long-Range (LoRa) Communication

To address the range limitation, the 433MHz RF modules could be replaced with **LoRa (Long Range)** modules (e.g., Semtech SX1278). LoRa is a low-power, wide-area-network (LPWAN) technology that operates on a similar frequency but can achieve a range of several *kilometers* instead of meters. This would allow a single central controller to cover a vast area, or even a small village co-op.

7.3.3. WSN Mesh Network Implementation

For truly massive farms, the system could be evolved from a "star" topology to a "mesh" topology (drawing on the research from [2]). In a mesh, nodes can relay messages for each other, allowing a signal to "hop" from the farthest node back to the central controller, effectively creating a network with unlimited range.

7.3.4. Integration with Active Fire Suppression Systems

The controller's digital output pins, which now trigger a buzzer, could be used to trigger a solenoid valve. This could automatically open a water line or start a water pump, transforming the system from a passive *detection* system to an active *suppression* system for small, nascent fires.

7.3.5. Data Logging and Post-Fire Analysis

A simple MicroSD card module could be added to the central controller. This would allow it to log all sensor readings, events, and alarms. This data would be invaluable for post-fire analysis, for fine-tuning the fusion algorithm's thresholds, and for insurance or-investigative purposes.

7.3.6. Smartphone Application Integration

While the system is "offline first," a future version could include an optional GPRS data connection. The controller could send its data to a simple cloud database, which would populate a smartphone app. This app could show sensor status, battery levels, and provide push notifications, adding a layer of modern convenience for tech-savvy farmers.

CHAPTER 8: CONCLUSION

8.1. Summary of Work

This project confronted the devastating, real-world problem of accidental crop fires, which threaten the livelihoods of India's most vulnerable small and marginal farmers. Recognizing that existing high-tech solutions are rendered ineffective by the "digital divide" (i.e., poor internet and high cost), this project set out to design a solution from the ground up, built specifically for the constraints of a rural environment.

The result is the successful design, implementation, and validation of an "IoT-Based Farmer Field Fire Detection and Alert System." This system is a standalone, battery-powered, and offline-first solution. It uses a distributed network of wireless sensor nodes, placed at the corners of a field, to monitor for signs of a fire. These nodes report data to a secure, underground central controller, which acts as the system's brain.

The project's key innovation lies in its multi-sensor fusion algorithm, which intelligently combines data from flame, smoke, and temperature sensors to reliably detect a real fire while rejecting common false alarms like sunlight or stubble burning. Upon valid detection, the system provides an immediate local alarm and, most critically, uses an integrated GPS and GSM module to send an instant SMS alert—complete with precise map coordinates—to the farmer and emergency services, all without requiring an internet connection.

8.2. Achievement of Objectives

As demonstrated by the rigorous field tests in Chapter 6, the project has successfully met all its primary objectives. The prototype proved its ability to detect a real fire in under 20 seconds and deliver an actionable SMS alert in under one minute. It simultaneously proved its intelligence by successfully rejecting false alarms from smoke and direct sunlight. The calculated 86-day battery life of the sensor nodes confirms the viability of the low-power design. The GPS-enabled SMS alert effectively solves the problem of rural location ambiguity. The system is affordable, scalable, and robust.

8.3. Final Remarks and Vision

This project delivers a complete, end-to-end solution that directly addresses the technological and economic barriers faced by rural farmers. It moves beyond a theoretical concept to a practical, field-tested, and deployable system. By prioritizing affordability, offline functionality, and reliability, this system provides a specialized, accessible safety net where conventional high-tech solutions fail. It offers an effective tool for protecting livelihoods and enhancing food security at the grassroots level.

The successful development of this prototype serves as a validated blueprint for large-scale implementation. The future vision is one where this technology, enhanced with solar power and LoRa communication, can be deployed not just on individual farms, but as communitywide or village-level alert networks. This project stands as a testament to the power of applying sound electronics and communication engineering principles to create appropriate, accessible, and high-impact solutions for real-world human problems.

REFERENCES

[1] P. S. Kshirsagar, S. R. Sutar, R. B. Wagh, and S. G. Nile, “Forest Fire Detection System using GPS & GSM,” Int. J. Adv. Res. Comput. Commun. Eng., vol. 5, no. 4, pp. 458–460, Apr. 2016.

[2] J. N. Al-Karaki and A. E. Kamal, “Routing techniques in wireless sensor networks: a survey,” IEEE Wireless Commun., vol. 11, no. 6, pp. 6–28, Dec. 2004, doi: 10.1109/MWC.2004.1368893.

[3] L. Hartung, J. Liddle, and M. Bues, “A low-cost IoT system for monitoring and early detection of wildfires,” HardwareX, vol. 8, Art. no. e00135, Oct. 2020, doi: 10.1016/j.ohx.2020.e00135.

[4] W. Li, X. He, L. Liu, and B. Li, “Multi-Sensor Data Fusion Algorithm for Fire Detection,” Procedia Eng., vol. 11, pp. 228–234, 2011, doi: 10.1016/j.proeng.2011.04.651.

[5] S. R. Nandurkar, V. R. Thool, and R. C. Thool, “IoT based Smart Agriculture System,” in Proc. Int. Conf. Comput. Commun. Informat. (ICCCI), Coimbatore, India, 2014, pp. 1–5, doi: 10.1109/ICCCI.2014.6921401.

[6] Arduino, “Arduino Nano,” Arduino Official Store, 2023. [Online]. Available: https://store.arduino.cc/products/arduino-nano. [Accessed: Aug. 20, 2025].

[7] SIMCom Wireless Solutions, SIM800L Hardware Design Guide, 2017.

[8] u-blox AG, NEO-6 Series Hardware Integration Manual, 2013.

[9] M. A. A. Al-Qurabat and A. J. H. Al-Hamdani, “A Review of Wireless Sensor Networks for Fire Detection Systems,” Int. J. Comput. Sci. Inf. Secur., vol. 14, no. 9, pp. 786–793, 2016.

[10] Adafruit, “DHT11, DHT22, and AM2302 Sensors,” Adafruit Learning System, 2021. [Online]. Available: https://learn.adafruit.com/dht. [Accessed: Sep. 10, 2025].

[11] M. Hart, “TinyGPS++ Library,” Arduiniana, 2023. [Online]. Available: http://arduiniana.org/libraries/tinygpsplus/. [Accessed: Sep. 20, 2025].

[12] R. C. González, P. W. Wong, A. Martínez, and S. Rajasekaran, “A Low-Cost Multi-Sensor System for Early Fire Detection in Outdoor Environments,” Sensors, vol. 18, no. 4, Art. no. 1063, 2018, doi: 10.3390/s18041063.